Quantum Computing for Neutrino–Nucleus Scattering

What one neutrino–nucleus calculation costs on a fault-tolerant quantum computer

Gabriel Perdue

Fermi National Accelerator Laboratory

Neutrino school — 13 November 2026

Plan of the lecture (two hours, with a short break in the middle)

Hour 1: the problem, the machine, the target, the budget

1. Why quantum computing for neutrino physics?
2. A quantum computer, for a physicist (the on-ramp)
3. The target: the neutrino response of carbon on a lattice
4. The 2033 machine, and the budget we measure against

A short break.

Hour 2: one calculation, costed piece by piece

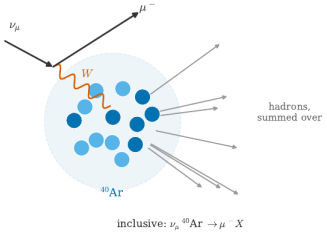
5. Carbon on qubits
6. Measuring an energy on a quantum computer
7. The cost, piece by piece
8. Closing matter: argon, the roadmap, exercises

What to expect. Hour 1 assumes quantum mechanics and DUNE, not quantum computing. Hour 2 prices one calculation, the inclusive neutrino response of $^{12}\mathrm{C}$ on a 4^3 lattice. A companion notebook works each of its steps in full (the code slide).

Expect a challenge! The notebook will take weeks to fully absorb. Skim the slides for a flavor if that's all you want, and dig into the notebook and code if you want to go *deep*. If you make it through the notebook and want to write a paper on any of the problems, reach out to me.

DUNE's precision program meets a theory wall

- The Deep Underground Neutrino Experiment (DUNE) targets leptonic CP violation and the mass ordering through $\nu_\mu \leftrightarrow \nu_e$ oscillations at few-GeV beam energy, on a liquid-$^{40}\mathrm{Ar}$ target.
- DUNE's Technical Design Report (TDR) budgets $\sim$2% post-near-detector signal-normalization uncertainty on the appearance sample [TDR Vol. II]; once flux and detector are constrained, the residual is dominated by cross-section / nuclear-model systematics.
- Generator-level model spread (GENIE vs. GiBUU under the DUNE flux) shifts projected CP-violation sensitivity by up to $\sim 1\sigma$ and mass-ordering sensitivity by up to ~ 2–3σ [Nagu et al. 2019].
- At statistics-limited $\sqrt{\text{exposure}}$ scaling — our conversion — a 0.5–1σ loss at the 5σ frontier is equivalent to a $\sim$25–50% discovery-exposure penalty.



"Inclusive": any hadronic final state.

Figure: H. Lamm (Fermilab), 2026.

So: the inclusive neutrino–nucleus response is the leading remaining *theory* systematic on a CP-violation measurement.

Classical state of the art: carbon is not easy

Event generators (GENIE, GiBUU, NuWro)

- Phenomenological models with cascades: tunable, fast, full final states; uncontrolled at the few-percent level exactly where DUNE needs them.

Ab initio, at its best: carbon

- Green's-function Monte Carlo has computed $^{12}\mathrm{C}$'s inclusive electroweak responses with one- and two-body currents [Lovato et al., arXiv:1605.00248, arXiv:2003.07710].
- The weak-response campaign took about 1.3×10^8 core-hours on the Mira supercomputer; inclusive only; beyond $A \approx 12$, "out of the question" by its authors' account.

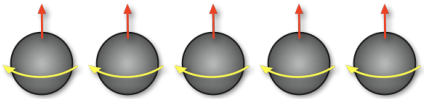
The two walls

- Real time is blocked. $\langle \Psi_0 | J^\mu e^{-iH\tau} J^\nu | \Psi_0 \rangle$ has a fermion sign problem; classical work stays in imaginary time, and inverting imaginary-time data above the quasi-elastic peak is ill-posed.
- The Hilbert space explodes. Twelve nucleons on the 64-site lattice of hour 2: $\binom{64}{3}^4 \approx 3 \times 10^{18}$ amplitudes (counted later this hour).

"Classically not easy" is the claim of this lecture: carbon's response is a supercomputer campaign today, reached only through an inversion. The quantum method is built to walk around both walls.

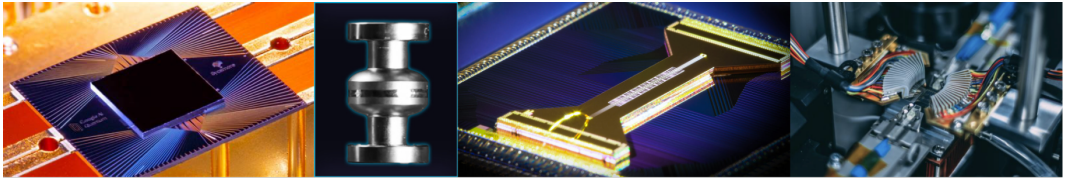
Zoom out briefly: A computer is a physical system you push around

- **Computation** is manipulating a physical system by rules and interpreting the result. An abacus adds without knowing it; a transistor adds with voltages. The rules are the algorithm; the hardware is the simulator.
- So the design question is which physics you build the computer out of. A transistor is a two-level system that is cheap, fast and classical. A spin is a two-level system that is quantum.
- Quantum computers give us quantum physical primitives to express algorithms: entanglement, superposition, and interference.



A register: five two-level systems, each a spin.

What a quantum computer does differently

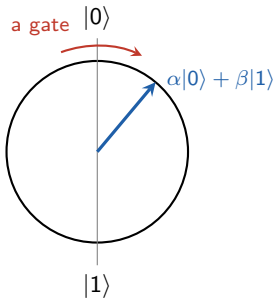


Google's Sycamore transmon chip; a Fermilab SQMS superconducting cavity; a Quantinuum ion trap; a Xanadu photonic chip (press images).

- It is not "many calculations at once." A register of n qubits holds one state in a 2^n-dimensional space. Each gate moves the whole amplitude distribution, and a measurement returns one *sample* from it.
- The native application is simulating quantum systems. Feynman's proposal and Lloyd's construction [Feynman, Int. J. Theor. Phys. 21, 467 (1982); Lloyd, Science 273, 1073 (1996)].
- **Both walls fall.** The register is linear in the size of the lattice, and real time is native: unitary evolution has no sign problem. That is the whole case for the rest of today.

The gate model (i): qubits, states, measurement

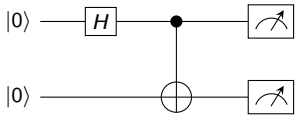
- **Qubit:** any two-level system. Its state is $\alpha|0\rangle + \beta|1\rangle$ with complex amplitudes, $|\alpha|^2 + |\beta|^2 = 1$. **Superposition** means both amplitudes are nonzero; the relative **phase** between them is physical, and it is what interferes.
- **Register:** n qubits, one state of 2^n amplitudes (the bookkeeping wall: back-up 5).
- **Entanglement:** a register state that is not a product of one-qubit states. The Bell pair $(|00\rangle + |11\rangle)/\sqrt{2}$ is the smallest.
- **Measurement:** read the register in a basis. You get one bit string, with probability $|\text{amplitude}|^2$, and the state collapses to it. Repeat for statistics: a quantum computer's output is always samples.



A one-qubit state as a direction (one angle drawn; the phase is the second). A gate rotates it.

The gate model (ii): gates, circuits, the T gate

- **Gate:** a unitary on one or two (usually) qubits. A one-qubit gate rotates the arrow of the previous slide: X flips $|0\rangle \leftrightarrow |1\rangle$, H makes the equal superposition, $R_z(\theta)$ turns the phase by θ.
- **CNOT:** flips the target when the control is $|1\rangle$; the two-qubit gate that makes entanglement. H then CNOT turns $|00\rangle$ into the Bell pair (figure).
- **Circuit:** each wire is a qubit, time runs left to right, boxes are gates, the meter is a measurement.
- **Universal gate set:** $\{H, S, \mathrm{CNOT}, T\}$ approximates any unitary to any precision [Nielsen & Chuang, Ch. 4], with $T = R_z(\pi/4)$. Every "T count" in hour 2 is this gate: on a fault-tolerant machine it is the expensive one (the fault-tolerance slide).



Two gates make a Bell pair; the meters read both qubits.

The gate model (iii): the gates as matrices

With $|0\rangle = \binom{1}{0}$ and $|1\rangle = \binom{0}{1}$, a gate is a unitary matrix and applying it is matrix multiplication. Each column says where one basis state goes.

The Pauli matrices, which you know:

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

The three new ones:

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}, \quad T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}.$$

- H, the Hadamard gate, makes the equal superpositions: $|0\rangle \to (|0\rangle + |1\rangle)/\sqrt{2}$ and $|1\rangle \to (|0\rangle - |1\rangle)/\sqrt{2}$.
- S and T only put a phase on $|1\rangle$, and they are square roots in a chain: $T^2 = S$ and $S^2 = Z$.
- $R_z(\theta) = e^{-i\theta Z/2}$ puts $e^{\mp i\theta/2}$ on $|0\rangle$ and $|1\rangle$, so T is $R_z(\pi/4)$ up to an overall phase, which no measurement can see.

Two qubits have four amplitudes, in the order $|00\rangle$, $|01\rangle$, $|10\rangle$, $|11\rangle$, with the control on the left:

$$\mathrm{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

- The top-left block is the identity: control $|0\rangle$, target left alone. The bottom-right block is X: control $|1\rangle$, target flipped.

The gate model (iv): the Bell pair, multiplied out

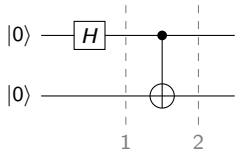
Start: $|00\rangle = (1,0,0,0)^{\mathsf{T}}$.

Step 1, H on the first qubit and nothing on the second. That 4-by-4 matrix is $H \otimes I$, each entry of H times the 2-by-2 identity:

$$(H \otimes I)\,|00\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix} = \frac{|00\rangle + |10\rangle}{\sqrt{2}}.$$

Step 2, the CNOT swaps the last two amplitudes:

$$\mathrm{CNOT}\, \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix} = \frac{|00\rangle + |11\rangle}{\sqrt{2}}.$$



- The circuit runs left to right and the product right to left: the whole circuit is the one matrix $\mathrm{CNOT} \cdot (H \otimes I)$.
- Measure both qubits: 00 or 11, each with probability $1/2$.

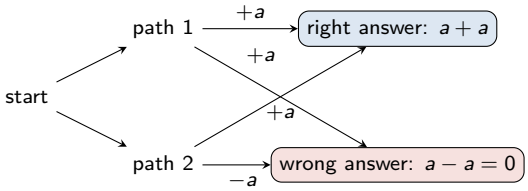
This state is entangled: $(a|0\rangle + b|1\rangle)(c|0\rangle + d|1\rangle)$ would need ac, $bd \neq 0$ and $ad = bc = 0$, which cannot all hold.

The gate model (v): Clifford gates, the easy half

- **Pauli operators** are the one-qubit X, Y and Z, and their products across a register, *Pauli strings* such as $X \otimes Z \otimes Z \otimes X$. Hour 2 writes the nuclear Hamiltonian as a sum of Pauli strings.
- **Clifford gates** turn every Pauli string into another Pauli string: $HXH = Z$, $SXS^\dagger = Y$, and CNOT copies an X on the control onto the target. H, S and CNOT generate them all.
- **The Gottesman–Knill theorem:** a circuit of Clifford gates, started in $|0\cdots0\rangle$ and read by measuring Pauli operators, can be simulated efficiently on a classical computer, by tracking how each Pauli string moves [Gottesman, arXiv:quant-ph/9807006]. Clifford circuits make plenty of entanglement, so entanglement alone is not the speedup.
- The T gate is where the quantum advantage enters. T is not a Clifford gate ($TXT^\dagger = (X+Y)/\sqrt{2}$, not a Pauli string). On a fault-tolerant machine the same split sets the prices: the Cliffords are cheap and T is expensive (the fault-tolerance slides).

Where advantage comes from (i): interference, not parallelism

- A classical randomized computer adds *probabilities*: never negative, so paths to a wrong answer never cancel.
- A quantum computer adds *amplitudes*, with phases. Paths to a wrong answer can cancel; paths to the right one can add.
- An algorithm is a choreography of interference: arrange, with polynomially many gates, that the wrong answers cancel *before* the one measurement.
- This needs structure. The answer must be a global property that interference can read: a period, a phase, an eigenvalue. Most problems have none, and get no speedup.



The families that have the structure: period finding (Shor), amplitude amplification (Grover, which returns in hour 2), and simulating quantum dynamics (next slide).

Where advantage comes from (ii): the physics case

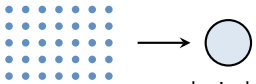
- Simulating quantum dynamics is the speedup with a theorem. For a Hamiltonian of local terms the dynamics e^{-iHt} costs polynomially many gates [Lloyd, Science 273, 1073 (1996)], where a classical computer pays 2^n.
- **What you can ask of it.** Not the wavefunction: 2^n amplitudes cannot be read out. One number at a time: an energy. *Phase estimation* turns the dynamics into a phase on a few extra qubits and reads that phase as a bit string.
- **What comes out.** Each run returns one energy, sampled with the weight the physics gives it. A response function is a histogram of such samples, so the output costs shots, and the shot count is part of the price.
- **What the theorem does not cover.** Preparing the input state, here the nucleus's ground state. Hour 2's open problem sits exactly there.

NISQ: the analog way

- **Today's machines:** 10^2 to 10^3 physical qubits. A gate is a pulse that rotates the state by an angle set by the pulse length. The angle is continuous, and so is the error.
- **Every gate errs at the 10^{-3} to 10^{-2} level, and nothing corrects it.** Errors accumulate with depth, so a program is limited to about a thousand gates before the signal is gone.
- **The output** is an expectation value estimated from many shots, with error *mitigation* that reduces bias and corrects nothing.
- Analog in the sense that matters: the precision is set by the hardware's noise and cannot be bought by running longer. The variational algorithms (VQE and relatives) you may have heard of live here [Preskill, arXiv:1801.00862].

Fault tolerance: the digital way

- **Encode** each logical qubit in many physical ones, check parities continuously, correct as the program runs. The logical error rate per operation, ϵ_l, falls exponentially with the code's size (next slide).
- **Compile** everything to a discrete gate set, Clifford + T. No continuous angles remain: a rotation $R_z(\theta)$ becomes a sequence of about 34 T gates (hour 2 prices this).
- Digital: arbitrarily long computations at a price per gate, like a classical computer.
- **The price.** Clifford gates are cheap; each T gate consumes a *magic state* distilled by factories that take much of the machine.



$\sim 10^3$ physical qubits

$R_z(\theta) \;\rightarrow\; T\,H\,T\,S\,T\cdots$ (≈ 34 T gates)

The three currencies of hour 2. Logical qubits (the register); T gates per circuit (the work); ϵ_l per gate (the reliability). Wall time $\approx$ T count / T rate.

Error correction in one slide

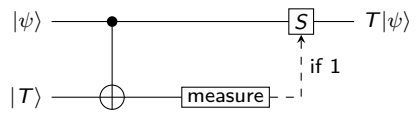
- **Why it is hard:** you cannot copy a qubit, you cannot look at it without collapsing it, and its errors are continuous, so majority voting is out.
- **The trick:** measure *parities* of groups of qubits. The pattern of outcomes, the syndrome, locates an error without revealing the state, and the measurement itself snaps a continuous wobble onto a discrete, correctable flip. The surface code does this on a 2D grid; its distance d is the number of physical errors needed to fool it.
- **What it costs (schematic):** the logical error rate falls as $\epsilon_l \sim (p/p_{\mathrm{th}})^{(d+1)/2}$. At a physical rate p ten times below threshold, the $\epsilon_l \sim 10^{-10}$ that hour 2's circuit needs takes $d \approx 19$: $\sim 2d^2 \approx 700$ physical qubits per logical one, a million for a thousand logical qubits. Google's 2024 experiment was the first to show the scaling on hardware [arXiv:2408.13687].
- Study it later, as a rite of passage: the primer starting at back-up 27 and the two papers on the last slide.

Fault-tolerant gates (i): the Clifford gates are nearly free

- **Pauli gates are never run.** An X or a Z only relabels later outcomes, so the control software keeps a classical record, the *Pauli frame*, and reinterprets measurement results by it.
- **The other Clifford gates can be moved out of the way.** Push each one through the circuit toward the end. A T gate it passes turns into a $\pi/8$ rotation about a longer Pauli string, still one T-type operation; at the end the Cliffords fold into the final measurements [Litinski, arXiv:1808.02892].
- **What is left to run:** one $\pi/8$ rotation for each T gate, made by consuming a magic state (next slide), and measurements.
- That is why every cost in hour 2 is a T count. A Toffoli (we'll meet it soon) is not a Clifford gate, so it is priced in T gates too.

Fault-tolerant gates (ii): magic states, or how a T gate is made

- **A T gate by teleportation.** Prepare the one-qubit state $|T\rangle = T|+\rangle = (|0\rangle + e^{i\pi/4}|1\rangle)/\sqrt{2}$. A CNOT and a measurement move the T onto your qubit; when the measurement reads 1, an S gate, a Clifford, finishes the job (figure).
- So the hard part is a supply of very clean $|T\rangle$ states. Bravyi and Kitaev named them *magic states*: it is "a rather surprising coincidence that one and the same state" both makes the Clifford gates universal and can be purified by Clifford operations alone [Bravyi & Kitaev, PRA 71, 022316 (2005)].
- **Distillation:** 15 noisy copies in, one cleaner copy out, with error $\epsilon_{\text{out}} \approx 35\,\epsilon^3$ [ibid., Eq. 38].



The T gate costs one magic state; everything else in the circuit is Clifford. Multiplied out on back-up 6.

Fault-tolerant gates (iii): T factories set the machine's speed

- **A factory** is a region of the chip that runs distillation without stopping: noisy magic states in, clean ones out, one every so many error-correction cycles. Each T gate occupies many physical qubits for a while (the arithmetic of distillation: back-up 7).
- **The T rate** is the number of factories times their output. A computation is a queue of T gates waiting on the factories, so wall time $\approx$ T count / T rate. Hour 2 assumes 10^6 T per second on one machine. That is our assumption: no factory has been built at anything like these rates. For scale, one early, speculative layout would run at 7×10^3 with one factory and about 8×10^4 with eleven, on 55,000 and 120,000 physical qubits, taking a physical error of 10^{-4} and a 1 μs error-correction cycle [Litinski, arXiv:1808.02892].
- **Factories take room on the chip.** The qubit counts of hour 2 are the qubits the algorithm names, and the factories come on top.

Where the software is: gates are the assembly language

Today

- A program is a list of gates on named qubits. The Bell pair in OpenQASM is five lines:

```
OPENQASM 2.0; include "qelib1.inc";
qreg q[2]; creg c[2];
h q[0];
cx q[0],q[1];
measure q -> c;
```

- Simulators and today's devices, 10^2 to 10^3 noisy physical qubits, run exactly this.

A fault-tolerant application

- 10^9 hard operations (T gates and Toffolis) in the 2033 budget, with many more easy (Clifford) gates around them, so the full gate list is longer still. Nobody writes it out, and no simulator runs it.
- The program is a *tree of subroutines*: time evolution, block-encodings, phase estimation, arithmetic, each with a cost formula attached. Estimates are computed on the tree, never on a circuit [Qualtran, arXiv:2409.04643]. Every number in hour 2 is built this way.
- Below the tree, a logical compiler must turn Clifford+T into surface-code operations and feed magic states at rate [Litinski, arXiv:1808.02892]. That layer is research-grade today; nothing has run end to end at 10^9.

The gap between the two columns is where the software field is.

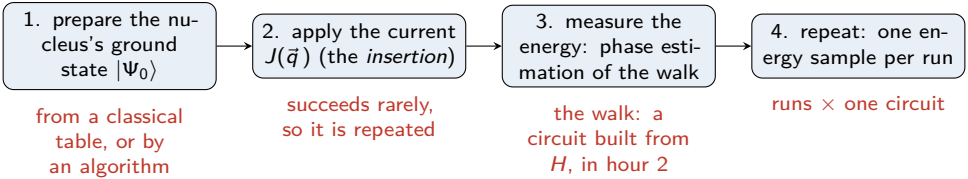
The object: the response is a histogram of final-state energies

For $\nu_\mu + A \to \mu^- + X$ at four-momentum transfer $q = (q_0, \vec{q}\,)$:

$$W^{\mu\nu}(q_0, \vec{q}\,) \;=\; \sum_f \langle \Psi_0 | J^{\mu\dagger}(\vec{q}\,) | f \rangle \langle f | J^{\nu}(\vec{q}\,) | \Psi_0 \rangle \, \delta(E_f - E_0 - q_0)$$

- **One histogram per component.** With $\mu = \nu$ the sum has only positive weights, $|\langle f | J^\mu | \Psi_0 \rangle|^2$: one number at each q_0. Five such responses make the cross section (back-up 8).
- **Read each one as a histogram.** Apply the current to the ground state; the result is a superposition of final states $|f\rangle$ with those weights. The response at energy transfer q_0 is the total weight at energy $E_0 + q_0$. The spread is the physics.
- **The final states are eigenstates of the full nuclear** H, so final-state interactions are in them by construction: no cascade model.
- A quantum computer samples the histogram directly: prepare $|\Psi_0\rangle$, apply J, measure the energy, repeat, with no inversion step.

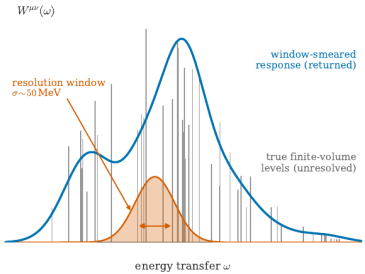
The recipe in four steps



- One run of steps 1–3 is *one circuit*; the 2033 budget of the next section limits its T gates and its qubits. Step 4 multiplies circuits into wall time.
- Step 2 succeeds only with a small probability, so the circuit repeats steps 1–2 inside an amplification loop: the ground state is prepared tens of times per circuit. *Of what we can price for carbon, about* 99% *is step 3. Step 1 is not priced yet, and it may cost even more.*
- Hour 2 prices steps 2 and 3 for $^{12}\mathrm{C}$ on the 4^3 lattice, and what step 1 may spend, from our tested code and four papers: the Hamiltonian [arXiv:1911.06368], the walk's circuits [Babbush et al., arXiv:1805.03662], the ground-state load [Fomichev et al., arXiv:2310.18410], the insertion [Roggero, Gu, Baroni & Papenbrock, arXiv:2009.13485].

What "computed" will mean here: carbon on the 4^3 lattice

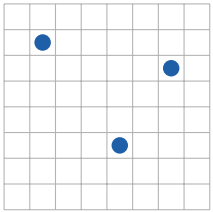
- **Model:** leading-order pionless effective field theory (EFT) on a cubic lattice: spacing 1.4 fm, 4^3 sites, a periodic box 5.6 fm on a side [arXiv:1911.06368]; no pions, contact interactions only.
- **Kinematics:** momentum transfers $|\vec{q}\,| \approx 0.22$–0.77 GeV, the ones this box allows; five response functions (back-up 8), each its own calculation.
- **Resolution:** energies read in 33.2 MeV bins, delivering 23.5 MeV RMS (hour 2 explains both); the target, 5% per bin at the peak, sets the number of runs.
- **What it is not:** exclusive channels, pion production, two-body currents. The box is also small: 5.6 fm is about the size of the nucleus, so the answer still depends on it. A production calculation enlarges the box until the answer stops changing.



The width is the deliverable. Figure: H. Lamm (Fermilab), 2026.

The nuclear wall, counted: carbon's 3×10^{18} amplitudes

- **The lattice of hour 2** has 64 sites ($4 \times 4 \times 4$). A site can hold one nucleon of each of four kinds, the *species*: proton or neutron, spin up or down ($p\uparrow$, $p\downarrow$, $n\uparrow$, $n\downarrow$). One site with one species is a *mode*: $64 \times 4 = 256$ modes.
- **Carbon** is twelve nucleons, six protons and six neutrons. The model conserves the number of each species, and carbon's ground state has three of each.
- **Pauli exclusion:** two nucleons of one species cannot share a site, and identical nucleons cannot be told apart. So a configuration of one species is a choice of 3 occupied sites out of 64: $\binom{64}{3} = 41{,}664$ ways.
- **The four species choose independently:** $41{,}664^4 \approx 3.0 \times 10^{18}$ configurations of all twelve. The quantum state has one amplitude for each configuration.



The 64 sites laid flat: one species places its three nucleons, in one of $\binom{64}{3}$ ways.

At one byte per amplitude that is 3×10^{18} bytes: three thousand world-class supercomputers, at a petabyte of memory each, for one state of one nucleus.

“Classically not easy”

- **The size.** The 3×10^{18} amplitudes counted in “The nuclear wall”. No laptop; and the ground state alone is beyond any classical table (hour 2 returns to this).
- **The best classical result.** GFMC’s $^{12}\mathrm{C}$ responses [arXiv:2003.07710]: about 1.3×10^{8} core-hours on Mira; inclusive; imaginary time, inverted by maximum entropy; richer currents than ours (two-body); by its authors’ account, closed beyond $A \approx 12$ and the Δ region.
- What the quantum method adds: real-energy sampling with no inversion; a systematic error that improves with EFT order; a path to heavier nuclei (argon, in the closing matter).
- What it does not add, yet: a smaller error bar on the same quantity. The leading-order Hamiltonian is cruder than the classical one; the claim is a different method, and a better number comes later if at all.
- **Why carbon:** the scintillator nucleus of the US program and a well-measured cross-section target.

DARPA’s Quantum Benchmarking Initiative says *utility-scale* means computational value exceeding computational cost: *keep asking which side of that line each estimate sits on*.

The 2033 machine: the budget we measure against

The DOE Office of Science 2026 request for information (RFI) on fault-tolerant systems describes two configurations [DOE RFI 2026]. The second is the budget every cost in hour 2 is measured against:

	logical qubits	hard ops per circuit (T or Toffoli)	logical error rate
2028 configuration	150–250	$\leq 10^5$	$\epsilon_l \leq 10^{-8}$ per gate
2033 configuration	$\gtrsim 10^3$	$\sim 10^9$	*none stated*

- The missing ϵ_l is set by the circuit. We budget 0.1 expected logical faults per circuit (10% of shots carry a fault), so $\epsilon_l \leq 0.1/N_T$: 10^{-10} per gate for a circuit that uses the whole budget. *The benchmark under-specifies; the application closes the spec.*
- Nothing in hour 2 fits the 2028 configuration; it returns in the closing matter.
- Fermilab submitted a whitepaper that, among other things, responds to this RFI. What follows is an independent calculation against the same RFI.

Hour 2 in one slide: where the T gates go

One circuit, for $^{12}\mathrm{C}$ on the 4^3 lattice:

$$T_{\mathrm{circuit}} = \underbrace{2^{W_q-1}\,(T_{\mathrm{prep}} + T_{\mathrm{select}} + T_{\mathrm{reflect}}) + T_{\mathrm{readout}}}_{\text{the energy measurement}} + \underbrace{(1+2m)\,(T_{\mathrm{load}} + T_{\mathrm{insert}}) + m\,T_{\mathrm{amp}}}_{\text{the ground state, loaded } 1+2m \text{ times}}$$

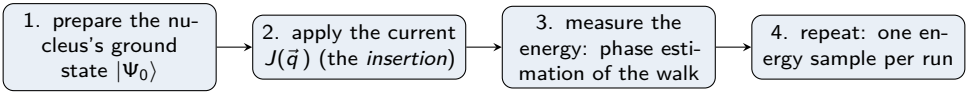
$$N_{\mathrm{qubits}} = \underbrace{4M}_{\text{the nucleus}} + \max(\text{the walk's ancillas},\ \text{the amplification's ancillas})$$

$$\text{wall time} = \text{shots} \times T_{\mathrm{circuit}} \,/\, (T \text{ rate})$$

- $M = 64$ sites; $W_q = 12$ energy bits; m amplification rounds, 20–29 here; *ancillas* are scratch qubits.
- Every symbol above gets its own slides and a number, in the plan's order: carbon on qubits (the $4M$), measuring an energy (the walk and the readout), then the cost piece by piece (the ground state, the qubits, the wall time).
- The closing matter turns the same checklist on published estimates, ours included.

Welcome back

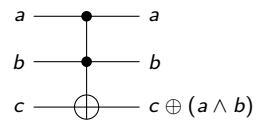
Hour 2: one calculation, the inclusive response of $^{12}\mathrm{C}$ on the 4^3 lattice, priced piece by piece against the 2033 budget. The recipe again:



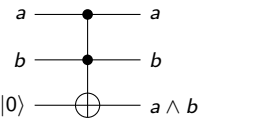
The order we take: the circuit words, then the nucleus on qubits; the block encoding, once; then the recipe's steps in the order the costs need: step 3, step 1, step 2; the sum; what the missing piece may cost; then errors, qubits, runs and the verdict. From the block encoding on, a tag at the top right of each slide names its step.

Circuit words for hour 2 (i): controlled gates, Toffoli, AND

- **Controlled gate:** apply a gate U to the target only when the control qubit is $|1\rangle$. CNOT is the controlled X.
- **Toffoli** (controlled-controlled-NOT): flips the target only when both controls are $|1\rangle$. It is not a Clifford gate; the textbook circuit uses 7 T gates.
- **AND:** a Toffoli onto a fresh $|0\rangle$ target writes $a \wedge b$ into it. Computing an AND costs 4 T, and erasing it later costs none [Gidney, Quantum 2, 74 (2018)].
- This lecture prices every Toffoli at 4 T, as Babbush et al. do [arXiv:1805.03662]: our Toffolis are ANDs that are erased afterward. The budget's wording could also let a Toffoli count as one hard operation if you had a native Toffoli.



Toffoli: 7 T (textbook)



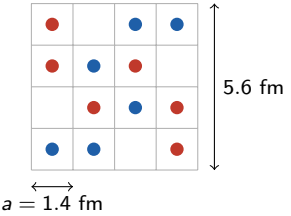
AND: 4 T, erased for free

Circuit words for hour 2 (ii): registers, chains, rotations

- **Register:** a named group of qubits that holds one quantity. Hour 2 has a *system register* (the nucleus, one qubit per lattice mode), an *energy register* (where phase estimation writes the energy), and *ancilla registers* (scratch space).
- **Ancillas start and end in $|0\rangle$.** An ancilla left changed stays entangled with the system and spoils its state. Running a computation backward to clear its ancillas is *uncomputing*.
- **AND chain:** to test whether k qubits are all $|1\rangle$, compute the AND of the first two, then the AND of that and the third, and so on: $k-1$ ANDs, each into a fresh ancilla. Hour 2's reflections are AND chains.
- **Rotation $R_z(\theta)$:** a phase turn by an arbitrary angle. A fault-tolerant machine approximates it with a few dozen T gates (34 here, priced later).

The model: nucleons on a lattice

- **Leading-order pionless EFT** [arXiv:1911.06368]: nucleons hop between neighboring sites (the kinetic term) and interact only when they sit on the same site (two- and three-body contact terms). Spacing $a = 1.4$ fm; $4^3 = 64$ sites; a periodic 5.6 fm box.
- **Three numbers set it** at this spacing: the hopping energy $t = \hbar^2/(2M_N a^2) = 10.58$ MeV, the two-body contact $C_0 = -98.23$ MeV (attractive) and the three-body contact $D_0 = +127.84$ MeV (repulsive) [arXiv:1911.06368, Tab. I].
- The contacts' values come from a fit on a richer lattice Hamiltonian [Rokash et al., arXiv:1308.3386]. On this simpler one the deuteron binds at 4.3 MeV, not 2.2, so read it as a model with realistic scales.
- Each site hosts the four species $p\uparrow$, $p\downarrow$, $n\uparrow$, $n\downarrow$; carbon is three of each.



One slice of the lattice: twelve nucleons, protons in blue, neutrons in red.

The Hamiltonian, written out

With $a^\dagger_{i,f}$ creating a nucleon of species f on site i, $a_{i,f}$ removing one, and $n_{i,f} = a^\dagger_{i,f} a_{i,f}$ counting it (0 or 1) [arXiv:1911.06368, Eq. 1]:

$$H = 6tA - t \sum_f \sum_{\langle i,j \rangle} \left(a^\dagger_{i,f} a_{j,f} + a^\dagger_{j,f} a_{i,f} \right) + C_0 \sum_i \sum_{f<f'} n_{i,f}\, n_{i,f'} + D_0 \sum_i \sum_{f<f'<f''} n_{i,f}\, n_{i,f'}\, n_{i,f''}$$

- **Hopping** $(-t)$: a nucleon moves to a neighboring site $\langle i, j \rangle$ and keeps its species. With the constant $6tA$ ($A = 12$ nucleons) it is the kinetic energy on the lattice.
- **The two contacts only count:** C_0 for every pair of nucleons on one site (6 pairs of species), D_0 for every triple (4 triples).
- **One site, filled up:** two nucleons give $C_0 = -98$ MeV, three $3C_0 + D_0 = -167$ MeV, four $6C_0 + 4D_0 = -78$ MeV. Without the repulsive D_0, four would give $6C_0 = -589$ MeV: the three-body term stops nuclei from collapsing onto single sites.
- H conserves species number.

Two ways to put nucleons on qubits

A many-fermion state can be stored in two ways — two "*quantizations*":

Occupations (second quantization). Record, for each mode, whether it is empty or full: one qubit per mode, $|1\rangle$ occupied, $|0\rangle$ empty. 256 qubits.

- A basis state is a string of 256 bits with twelve 1's, three in each species' block of 64.
- Fermion signs live in the operators $a^\dagger$ and a; the Jordan–Wigner strings carry them.

Positions (first quantization). Record, for each nucleon, which mode it is in. Naming one of 256 modes takes $\log_2 256 = 8$ bits. $12 \times 8 = 96$ qubits.

- A basis state is twelve 8-bit numbers, one per nucleon.
- Fermion signs live in the state: it must flip sign when two nucleons' numbers are swapped. The circuit builds that into its starting state, and H keeps it.

Both hold the same 3×10^{18} states of carbon. A small helium example, written out bit by bit, starts at back-up 9.

The register: one qubit per mode

- **Our choice: occupations,** that is, second quantization. H is written in $a^\dagger$, a and n, so each of its terms acts on the qubits of the modes it names.
- **The register is the list of modes:** $64 \times 4 = 256$ qubits.
- **It holds more than carbon:** $2^{256} \approx 10^{77}$ basis states, for any number of nucleons. Carbon's are $\binom{64}{3}^4 \approx 3 \times 10^{18}$ of them, and H never leaves those, since it keeps each species' number.

encoding	qubits	grows with
occupations	256	the box
positions	96	the nucleon count

Positions take fewer qubits but costlier gates for the walk; they are where argon lives (closing matter).

"Which is better?" is the wrong question: which one wins depends on what the budget counts (back-up 12).

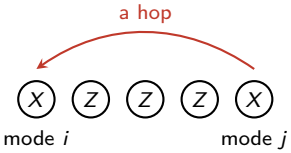
From fermions to qubits: the Jordan–Wigner strings

In second quantization every term of H is built from $a^\dagger$, a and n. Jordan and Wigner write each of them as Pauli operators on the 256 qubits:

- **Number operators are easy.** The occupation of mode j is $n_j = (1 - Z_j)/2$: 0 on $|0\rangle$, 1 on $|1\rangle$. Every on-site contact term is a product of n's, so a product of Z's.
- **Hopping needs signs.** Fermion operators anticommute: exchanging two nucleons flips the sign of the state. Operators on different qubits commute. Jordan and Wigner line the modes up in one order and attach a Z to every mode in between:

$$a_i^\dagger a_j + a_j^\dagger a_i = \tfrac{1}{2}\left(X_i\, Z \cdots Z\, X_j + Y_i\, Z \cdots Z\, Y_j\right).$$

- The Z's pick up one sign for each occupied mode the hop passes over. One hop is two Pauli strings, as long as the stretch of modes between its ends.



One of a hop's two strings.

On a 3D lattice, neighbors along y and z sit far apart in the line, so their strings are long. A string's length costs only Clifford gates.

One site's 38 terms, and $\Gamma = 2{,}432$

In Pauli strings the Hamiltonian is the same list of terms at every site [arXiv:1911.06368, App. C]:

- 24 **hopping strings:** one bond to the next site in each of 3 directions, $+x$, $+y$, $+z$ (a site's $-x$ bond is its neighbor's $+x$ bond), for each of 4 species. A bond's term moves a nucleon either way and makes 2 strings, $XZ \cdots ZX$ and $YZ \cdots ZY$: $3 \times 4 \times 2 = 24$.
- 14 **on-site strings,** the two contacts with each $n = (1 - Z)/2$ multiplied out: 4 single Z's, one per species; 6 pairs ZZ, one per pair of species; 4 triples ZZZ: $4 + 6 + 4 = 14$.
- 38 **per site, and** $\Gamma = 38 \times 64 = 2{,}432$ **terms** over the lattice, each a Pauli string U_γ with a coefficient.
- A constant is left over. It shifts every energy by the same amount, so it is dropped.

per site	strings	each with
hopping	24	$-t/2$
single Z	4	$-\frac{3}{4}(C_0 + D_0/2)$
pairs ZZ	6	$\frac{1}{4}(C_0 + D_0)$
triples ZZZ	4	$-D_0/8$
one site	38	
64 sites, Γ	2,432	

Two numbers from this list set every cost of the energy measurement: Γ, the number of terms, and λ, the total size of their coefficients.

λ: the size of the Hamiltonian, in one number

- **Definition.** Write $H = \sum_\gamma c_\gamma U_\gamma$, Pauli strings U_γ with real coefficients c_γ. Then $\lambda = \sum_\gamma |c_\gamma|$, the *one-norm* of the coefficients. No energy of H (less its constant) exceeds λ in size.
- **For our lattice**, site by site [our library: the terms of arXiv:1911.06368, Eq. 1, summed as in its Eq. C4]:

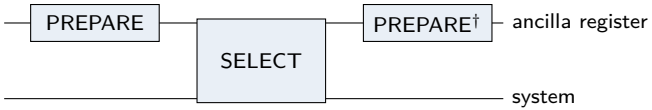
24 hopping strings, $\lvert t\rvert/2$ each: 12×10.58	127.0 MeV
14 on-site strings: $3\lvert C_0 + D_0/2\rvert + 1.5\lvert C_0 + D_0\rvert + 0.5\lvert D_0\rvert = 102.9 + 44.4 + 63.9$	211.3 MeV
one site	338.2 MeV
$\times$ 64 sites: λ	21,646 MeV

- The couplings are the model's three numbers [arXiv:1911.06368, Tab. I]: $t = 10.58$, $C_0 = -98.23$ and $D_0 = +127.84$ MeV.
- **Why it matters.** A quantum computer can only apply H/λ, inside a unitary, so λ sets the scale of every cost of the energy measurement.

Block encoding (i): applying H with unitaries

recipe steps 2, 3

- A quantum computer applies only unitaries, and H is not one. But $H = \sum_\gamma c_\gamma U_\gamma$ is a list of Pauli strings, and each string U_γ is a unitary the machine can apply.
- **The plan:** apply every string at once, in superposition, each weighted by its coefficient, with a few extra qubits, an *ancilla register*, holding the weights. Then recombine. This is a *linear combination of unitaries* [Babbush et al., arXiv:1805.03662].
- **The weights must be probabilities:** $p_\gamma = |c_\gamma|/\lambda$, which add up to one because λ is the one-norm. Each coefficient's sign is folded into its U_γ. So $H = \lambda \sum_\gamma p_\gamma U_\gamma$: H/λ is a weighted average of unitaries.
- **Three steps, each a unitary:** PREPARE writes the weights onto the ancillas, SELECT applies U_γ when the ancillas read γ, and PREPARE undone recombines.



Together they make one unitary, B, whose matrix holds H/λ in one block: a *block encoding* of H.

Block encoding (ii): PREPARE and SELECT

recipe steps 2, 3

- **PREPARE** writes the weights, as amplitudes, onto an ancilla register of about $\log_2 \Gamma$ qubits: $|0\rangle \to \sum_\gamma \sqrt{p_\gamma}\,|\gamma\rangle$. The amplitudes are square roots, so that reading the register would give γ with probability p_γ.
- **SELECT** applies U_γ to the system when the ancilla register reads $|\gamma\rangle$: $\sum_\gamma |\gamma\rangle\langle\gamma| \otimes U_\gamma$. On the superposition PREPARE made, it applies every string at once, each in its own branch.
- **Oracles.** PREPARE and SELECT are the block encoding's *oracles*: subroutines the algorithm calls and counts. Each is a large circuit (carbon's SELECT is a tree of 2,431 ANDs). The notebook builds them for one site, and our code counts their gates from the published constructions [Babbush et al., arXiv:1805.03662, Sec. III].

Block encoding (iii): following a state through them

recipe steps 2, 3

Follow $|0\rangle|\psi\rangle$ through PREPARE, SELECT and PREPARE undone:

$$
\begin{aligned}
|0\rangle|\psi\rangle &\xrightarrow{\text{PREPARE}} \sum_\gamma \sqrt{p_\gamma}\,|\gamma\rangle|\psi\rangle \\
&\xrightarrow{\text{SELECT}} \sum_\gamma \sqrt{p_\gamma}\,|\gamma\rangle\, U_\gamma|\psi\rangle \\
&\xrightarrow{\text{PREPARE}^\dagger} |0\rangle \sum_\gamma \sqrt{p_\gamma}\,\sqrt{p_\gamma}\, U_\gamma|\psi\rangle + s\,|\perp\rangle = |0\rangle\,(H/\lambda)|\psi\rangle + s\,|\perp\rangle.
\end{aligned}
$$

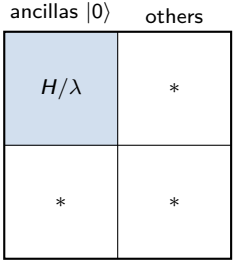
- After SELECT, each branch $|\gamma\rangle$ of the ancilla register holds a different state, $U_\gamma|\psi\rangle$. PREPARE undone sends each $|\gamma\rangle$ back to $|0\rangle$ with amplitude $\sqrt{p_\gamma}$, the amplitude PREPARE gave it, so each term carries $\sqrt{p_\gamma}$ twice.
- The branches return to $|0\rangle$ only as far as they agree. The part that returns is their weighted average, $(H/\lambda)|\psi\rangle$. The rest, $s\,|\perp\rangle$, is the *leftover*: $|\perp\rangle$ is a state of ancillas and system together, of length one, with the ancillas away from $|0\rangle$. Had every string turned $|\psi\rangle$ into the same state, all of it would return, with no leftover.

Block encoding (iv): a matrix hidden inside a unitary

- The three steps make one unitary, B. Sorted by the ancillas' state, its matrix holds H/λ in one block: hence the name.
- **The rule:** with the ancillas in $|0\rangle$ and the system in any state $|\psi\rangle$,

$$B\,|0\rangle|\psi\rangle = |0\rangle\,(H/\lambda)|\psi\rangle + s\,|\perp\rangle.$$

  B keeps lengths, so $\|H|\psi\rangle\|^2/\lambda^2 + s^2 = 1$: nothing vanishes, and the leftover carries the rest.
- **Measure the ancillas.** If every ancilla reads 0, the measurement has thrown the leftover away, and the system is left holding $H|\psi\rangle$, normalized. That happens with probability $\|H|\psi\rangle\|^2/\lambda^2$, so a large λ makes success rare.



Where the ancillas read 0 in and out, the block is H/λ; the block below it makes the leftover (the other blocks: back-up 15).

Block encoding's two uses: throw the leftover away, or keep it

recipe steps 2, 3

The recipe uses two block encodings, in opposite ways: the current's in step 2, and H's in step 3.

Step 2, the current J: throw it away

- J has a block encoding of its own. We want one state from it: $J|\Psi_0\rangle$, normalized. Phase estimation samples its energies, and their weights are the response.
- The leftover's energies are not the response, so it must go. The ancillas are measured once, before step 3, and the run is kept only if every one reads 0.
- Unaided, that happens in about one run in a thousand for carbon. Amplitude amplification first turns the state toward success (later slides).

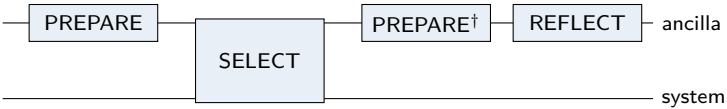
Step 3, the walk: keep it

- Phase estimation reads energies from a unitary whose eigenvalues carry them. For carbon it uses H's block encoding B $2^{11} = 2{,}048$ times, for 12 bits of energy (Phase estimation (ii)). If B were measured after each use, each use would succeed with probability $\|H|\psi\rangle\|^2/\lambda^2$, and over 2,048 uses the chances multiply to nothing.
- Unmeasured, B followed by REFLECT, a sign flip on the ancillas, is the walk: a unitary. Each step turns $|0\rangle|\psi_E\rangle$ by an angle set by E/λ (next slides).
- Nothing is thrown away, no step fails, and phase estimation reads the angle.

The walk: the block encoding, then one reflection

recipe step 3

One step of the walk acts on the system and the ancilla register:



- **PREPARE, SELECT, PREPARE†:** B, the block encoding of H.
- **REFLECT:** flip the sign of every ancilla state except $|0\rangle$, the one the block encoding starts from.
- **The walk** is this step repeated, *without* measuring the ancillas. Each step turns every energy eigenstate by the same angle, and the angle encodes the energy (the walk and phase estimation on one page: back-up 16).

PREPARE: the table of coefficients, from one site

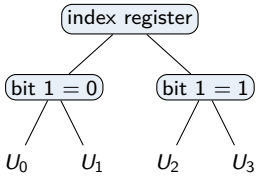
recipe step 3

- **The job:** turn $|0\rangle$ into $\sum_\gamma \sqrt{p_\gamma}\,|\gamma\rangle$ over all 2,432 terms.
- **The lattice shortcut.** Every site has the same 38 coefficients. So the ancilla register splits into a site index and a term index: an equal superposition over the 64 sites, times one site's 38 amplitudes [Babbush et al., arXiv:1805.03662, Figs. 11–12].
- **One site's 38 amplitudes by alias sampling** [ibid., Sec. III D]: an equal superposition, a table lookup and a comparison. The notebook works it through.
- **The price, per walk step:** 65 Toffolis and 2 rotations, and the same again to undo it: 656 T at 4 T per Toffoli and 34 per rotation.

SELECT: applying the term the ancilla register names

recipe step 3

- **The job:** for each γ, apply the Pauli string U_γ to the system when the ancilla register reads γ.
- **Unary iteration** [Babbush et al., Sec. III A]: visit the Γ index values in order. For each, a *flag* qubit reads $|1\rangle$ exactly when the index holds that value, and it controls U_γ. A tree of ANDs on the index qubits makes the flags (figure); neighboring values share the ANDs above where they part, so the tree costs $\Gamma - 1$ ANDs, one per node with two branches.
- A controlled Pauli string is Clifford (CNOTs and controlled Z's), so the strings are nearly free. The cost is the tree: $\Gamma - 1 = 2{,}431$ Toffolis, $9{,}724$ T.
- SELECT is 92% of a walk step, and it grows with the number of sites.



Each node with two branches is one AND on one bit of the index; each leaf fires once, and applies its string.

REFLECT: the sign flip that makes the walk

- **The job:** $R = 2|0\rangle\langle 0| - I$ on the ancilla register: leave the all-zero ancilla state alone, and flip the sign of every other.
- **How:** flip every ancilla qubit with X, so that "all zero" becomes "all one"; apply a Z conditioned on all of them, by one AND chain; undo the X's. In phase estimation the walk runs controlled on a bit of the energy register, which is one more control.
- **The price:** the ancilla register holds 43 qubits (the term index and the prepare's working qubits). With the control bit, the conditioned Z takes 42 ANDs: 168 T per step.
- Why it is there: repeating the block encoding alone builds up nothing useful (back-up 17). With the reflection between repeats, every step is the same rotation, and its angle is the energy.

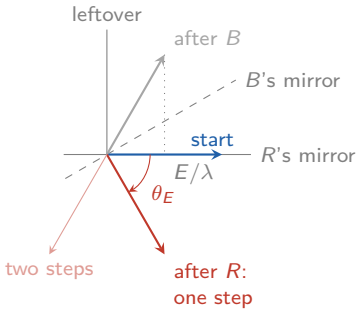
Why the walk encodes the energy: two mirrors make a rotation

recipe step 3

Take one energy eigenstate, $H|\psi_E\rangle = E|\psi_E\rangle$, and draw the plane of the start state $|0\rangle|\psi_E\rangle$ and its leftover $|\perp\rangle$:

1. **B tips the arrow.** $B|0\rangle|\psi_E\rangle = (E/\lambda)\,|0\rangle|\psi_E\rangle + s\,|\perp\rangle$: the arrow turns by θ_E, with $\cos\theta_E = E/\lambda$.
2. **Both boxes are mirrors.** $B^2 = I$ (every Pauli string squares to I), so B is a mirror, its line halfway between the start and where B sends it, at $\theta_E/2$. R keeps the start and flips the leftover: its line is the start axis. Neither leaves the plane [Low & Chuang, Quantum 3, 163 (2019)].
3. **One step.** After B the arrow is θ_E above the start axis; R puts it θ_E below. Two mirrors $\theta_E/2$ apart turn every arrow by θ_E: $W = RB$ is a rotation, eigenvalues $e^{\pm i\theta_E}$.
4. **Steps add.** B alone would bounce back; with R between, k steps turn by $k\theta_E$ [arXiv:1911.06368, Eqs. C5–C6, up to a fixed phase].

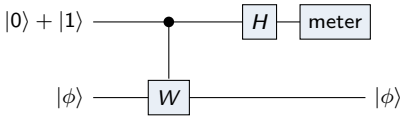
The energy is a rotation angle, exactly. The arrow has nowhere to arrive. It circles through the leftover and back, unmeasured, and phase estimation reads the angle.



Drawn for $\cos\theta_E = 0.5$, a turn of 60° per step. The notebook's pair state has $E/\lambda = -0.07$: a turn of 94°, nearly a quarter turn. Amplitude amplification, later, uses the same two mirrors to reach a target; the walk has none.

Phase estimation (i): one extra qubit reads a phase

- Suppose the system holds an eigenstate of the walk, $W|\phi\rangle = e^{i\theta}|\phi\rangle$. A phase on the whole state cannot be seen.
- **Make it relative.** Put a control qubit in $(|0\rangle + |1\rangle)/\sqrt{2}$ and apply W only when it is $|1\rangle$. The system is unchanged, and the control becomes $(|0\rangle + e^{i\theta}|1\rangle)/\sqrt{2}$: the phase has moved onto the control (*phase kickback*).
- A Hadamard and a measurement then return 0 with probability $(1 + \cos\theta)/2$: a little information about θ per run, for a number we want to 12 bits.
- **More bits at once:** further control qubits drive W^2, W^4, ..., so each holds the phase doubled once more.



One control qubit, one controlled walk.

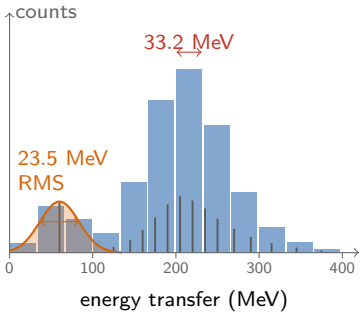
Phase estimation (ii): W_q bits, and what each one costs

recipe step 3

- **The energy register:** W_q control qubits, bit k driving W^{2^k}. An inverse quantum Fourier transform then turns their joint state into the phase written in binary, to W_q bits.
- **The walk steps:** $1 + 2 + 4 + \cdots + 2^{W_q - 1} = 2^{W_q} - 1$ in the textbook version. Babbush et al. run the walk forward on one branch and backward on the other, which halves it to $2^{W_q - 1}$ [arXiv:1805.03662]. For $W_q = 12$: 2,048 steps.
- **The bins.** The register splits the circle into 2^{W_q} angles, and the angle is $\arccos(E/\lambda)$. So near the middle of the spectrum an energy bin is $2\pi\lambda/2^{W_q}$ wide: $2\pi \times 21{,}646/4{,}096 = 33.2$ MeV.
- One more bit halves the bin and doubles the walk. Resolution is paid for in walk steps, in one circuit (why that is the best one can do: back-up 18).

Phase estimation (iii): what one run returns

- **A superposition in, one energy out.** Once the current is applied, the system is a superposition of energy eigenstates with weights $|\langle f|J|\Psi_0\rangle|^2$. Phase estimation returns one eigenstate's energy bin, picked in proportion to that weight: one sample of the response.
- **The lineshape.** Each energy is sharp, but its reading has a width: the register sees only 2^{W_q} steps of the walk, a finite time, which blurs an energy by about one bin, with tails farther out. A *window state*, not an equal superposition, cuts the tails [Babbush et al., Eq. 17].
- **The delivered error**, 23.5 MeV RMS: the window's lineshape and the circuit's error budget, added in quadrature. That, and not the bin width, is the resolution we quote.
- **The histogram.** Over many runs the counts are the response in those bins.



Gray: the final states' energies, sharp. Orange: how one of them is read, a bell 23.5 MeV RMS wide. Blue: the 33.2 MeV bins the runs fill.

One walk step, priced

recipe step 3

Carbon on the 4^3 lattice, from the library's counts; 4 T per Toffoli [Babbush et al., arXiv:1805.03662], 34 T per rotation (the rotation slides):

piece	what it is	Toffolis	rotations	T
PREPARE, and its inverse	one site's 38 coefficients by alias sampling, spread over the 64 sites	2×65	2×2	656
SELECT	a tree of ANDs, one per branching: $\Gamma - 1$ for $\Gamma = 2{,}432$ terms	2,431		9,724
REFLECT	an AND chain for a Z on the 43 ancilla qubits and the control bit: $44 - 2$	42		168
one step				**10,548**

- *Notice that we never computed a circuit.* Each line is a counting argument on a published construction, and that is what a resource estimate is.

The walk and the readout, priced

recipe step 3

	T gates	share of the 10^9 budget
the walk: 2,048 steps $\times$ 10,548	2.16×10^7	2.2%
the readout: the inverse Fourier transform ($\leq$ 132 rotations) and the window state ($\leq$ 4,096)	1.4×10^5	0.014%
the energy measurement	$\mathbf{2.17 \times 10^7}$	**2.2%**

- That is the whole energy measurement for carbon at 23.5 MeV RMS. On the next lattice, 5^3: 8.26×10^7 T, 8.3%. The box drives the cost twice: λ nearly doubles, which takes one more bit and twice the steps, and Γ nearly doubles the cost of each step.
- If the ground state were free, carbon would fit the budget with room to spare. It is not free, and it is loaded tens of times per circuit.
- For scale: the same measurement for $^4\mathrm{He}$ on 3^3 is 5.32×10^6 T, 0.53%.

Rotations (i): every angle is built from T gates

recipe step 3

- **Where the rotations are:** the prepare's angles (four per walk step), the inverse Fourier transform and the window state: at most 12,420 rotations $R_z(\theta)$ in what is priced, at arbitrary angles. The ground state's preparation, unpriced, adds its own.
- **The fixed-sequence way.** A fault-tolerant machine has only Clifford gates and T, and a fixed sequence of them reaches only a finite set of unitaries, spread over every axis of rotation. Landing within ϵ of the $R_z(\theta)$ you want costs about $3\log_2(1/\epsilon)$ T [Ross & Selinger, arXiv:1403.2975].
- **The way we price, repeat until success** [Bocharov, Roetteler & Svore, arXiv:1404.5320]. Run a short two-qubit sequence on the target and an ancilla, then measure the ancilla. On one outcome the target has received a z-rotation, from a far denser set of angles for the same T count; on the other it has received a known error that a Clifford gate undoes, and the circuit tries again. Counting every try, it needs on average about $1.15\log_2(1/\epsilon)$ T, in the form of Watson et al. [arXiv:2312.05344, Eq. 80]:

```
def repeat_until_success_t_count(epsilon_synthesis: float) -> int:
    """Expected T-gates to synthesize one R_z to precision epsilon."""
    return ceil(1.15 * log2(2.0 / epsilon_synthesis) + 9.2)
```

Rotations (ii): the error budget sets 34 T per rotation

- **One synthesis error budget for the whole circuit.** We allow all its rotation approximations together to be off by at most 10^{-2}. This is a bias we choose to tolerate, not a failure of the machine.
- **How the errors add.** A synthesized rotation is off by a small amount in a fixed direction, the same in every run: a *coherent* error. Such errors can line up, so N of them add linearly, and the budget is shared evenly, $\epsilon = 10^{-2}/N$. For the $N = 12{,}420$ rotations priced, $\epsilon \approx 8 \times 10^{-7}$.
- **The price.** Repeat until success at that precision: 34 T per rotation.
- **A weak lever.** A decade either way in the budget moves the price by about four T, and rotations are about 2% of what is priced.
- A randomized alternative loosens ϵ but not the price (back-up 19).

Loading a known state (i): a table of amplitudes onto qubits

recipe step 1

Suppose a classical computer holds the ground state as a table: D configurations ν_i (bit strings, one bit per mode) and their amplitudes α_i, far fewer than the strings the register could hold. Fomichev et al. load $\sum_i \alpha_i|\nu_i\rangle$ onto the system register in six steps [arXiv:2310.18410, Sec. III]:

1. **The amplitudes, on a small register.** Number the rows and make $\sum_i \alpha_i|i\rangle$ on an *enumeration register* of $L = \lceil \log_2 D \rceil$ qubits, by rotations.
2. **The configurations, by a table lookup.** Beside each $|i\rangle$, write row i's configuration into the system register (a tree of ANDs like SELECT's, D Toffolis): now $\sum_i \alpha_i|\nu_i\rangle|i\rangle$.
3. **The first bit of a short identifier of ν_i:** the parity, the sum mod 2, of a chosen set of system bits, written into an *identification register* by one CNOT per chosen bit.
4. **The other $2L-2$ bits the same way,** $2L-1$ in all, the sets chosen classically so that no two of the D configurations share an identifier [ibid., Lemma III.1].
5. **The row number erased:** for each row, flip the enumeration register from $|i\rangle$ back to $|0\rangle$ when the identification register reads that row's identifier: one multi-controlled operation per row.
6. **The identification register uncomputed** by the same CNOTs run again (a CNOT undoes itself). Both ancilla registers end in $|0\rangle$; the system holds $\sum_i \alpha_i|\nu_i\rangle$ alone.

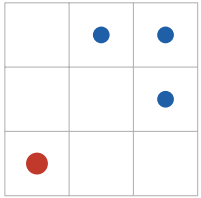
Loading a known state (ii): erasing the row number is the load

recipe step 1

- **After step 2 the system does not yet hold the state.** It holds $\sum_i \alpha_i |\nu_i\rangle|i\rangle$, each configuration tied to its row number. Anything acting on the system alone sees a *mixture*, as if one configuration had been drawn at random with probability $|\alpha_i|^2$. The interference between configurations that makes the sum an eigenstate is gone, and phase estimation would read a spread of energies in place of E_0.
- **Erasing it means a reverse lookup.** Step 2 run backwards erases the configurations, the wrong register. The erasure must go the other way: $|i\rangle$ back to $|0\rangle$ when the system register reads ν_i. Recognizing a whole configuration is an AND chain over every system qubit, once per row.
- **The identifier is the economy** (steps 3 and 4). Only D configurations are in play, so $2L-1$ parities of well-chosen bits already tell them apart, and parities cost CNOTs, not Toffolis. Step 5 then recognizes identifiers of $2L-1$ bits instead of whole configurations: $2L-2$ ANDs per row, $(2L-2)\,D$ in all.
- **The price** [arXiv:2310.18410, Eq. 17]: the amplitudes 2^{L+1} Toffolis (under $4D$), the lookup D, the erasure $(2L-2)\,D$, the CNOT steps nothing: fewer than $(2L+3)\,D$ Toffolis, most of them the erasure.

Loading a known state (iii): helium's trick, pin one nucleon

- **Helium-4 has exactly one nucleon of each species:** two protons and two neutrons, one spin up and one spin down of each.
- **Its ground state has zero total momentum,** so it is the sum, over the 27 translations of the 3^3 lattice, of the state with one chosen nucleon, say the $p\uparrow$, held at the origin.
- **That pinned state is small:** the other three nucleons on 27 sites each: $27^3 = 19{,}683$ configurations. A desktop computes the table.
- **Load it, then spread it** [our library]: a *shift register* in an equal superposition of the 27 translations; the lattice shifted by the amount it holds; then the shift register erased by reading where the $p\uparrow$ nucleon went. That last step works only because its species holds one nucleon. The spread costs 470 Toffolis and 6 rotations (204 T), beside 636,628 Toffolis for the load.



$p\uparrow$, pinned

One plane of the 3^3 lattice: the $p\uparrow$ nucleon held at the origin, the other three free to sit anywhere.

The ground state is the biggest cost: helium's load, priced

recipe step 1

^{4}He on the 3^3 lattice is the case where the ground state can be priced:

- **One load** of the 19,683-amplitude table, with the spreading: 2.55×10^6 T, almost all of it Toffolis.
- **Loaded** 29–43 **times per circuit** (the amplitude amplification slide says why): 0.74–1.1×10^8 T.

^{4}He, 3^3, 19.8 MeV RMS	T gates	share
the energy measurement	5.3×10^6	5–7% of the circuit
the ground state, 29–43 loads	0.74–1.1×10^8	93–95% of the circuit
one circuit	0.79–1.15×10^8	7.9–11.5% of the budget

The lesson: the walk is the cheap part. The state is the cost.

Carbon has no classical table

- **Helium's pinning fails.** Carbon has three nucleons of every species. Pin one $p\uparrow$ at the origin and spread: afterward the three $p\uparrow$ nucleons cannot tell which one was pinned, so the shift register cannot be erased, and it stays entangled with the system.
- **Pinning would not be enough anyway.** It divides the table by the 64 translations: $3 \times 10^{18}/64 \approx 5 \times 10^{16}$ amplitudes, still far beyond any classical table, and beyond any load circuit.
- GFMC's carbon ground state is a Monte Carlo object in the continuum, sampled and never tabulated. It does not help here.
- So the ground state must be prepared on the machine, by an algorithm rather than from a table. None has been priced for this system, by us or in the literature.

Applying the current: a block encoding that rarely succeeds

- **The current is not unitary.** Here $J = \rho_{\vec{q}} = \sum_x e^{i\vec{q}\cdot\vec{x}}\, n_x$, the proton density at momentum $\vec{q}$, one representative current (back-up 8): 128 terms of size $\frac{1}{2}$, two proton modes on each of the $M = 64$ sites. Applying it to the ground state is the *insertion*: $J|\Psi_0\rangle$, normalized, is the state whose energies E_f phase estimation reads; the histogram of $E_f - E_0$, the energy transfer, is the response.
- **It is block-encoded**, as H was: a small ancilla register (the term index: x, y, z and the spin, 7 qubits), PREPARE and SELECT. On this lattice every phase is a power of i, a Clifford gate on the index's bits, so the phases cost no T. Every ancilla reads 0, a *success*, with probability P_s, by the rule of Block encoding (iv) [Roggero, Gu, Baroni & Papenbrock, arXiv:2009.13485, Eq. 31]

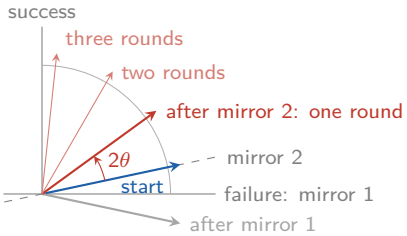
$$P_s = \frac{\eta^2}{\Lambda_J^2}, \qquad \eta^2 = \langle\Psi_0|J^\dagger J|\Psi_0\rangle \in [Z/2,\, Z] = [3, 6], \qquad \Lambda_J = 128 \times \tfrac{1}{2} = 64 = M.$$

- η^2 is the *strength*, the total weight of the response at this $\vec{q}$, every energy transfer added up. The nucleus fixes it, but for carbon we have no value: we take it between half and all of the proton count, the *strength band*, an estimate (back-up 20). Λ_J is the current's one-norm, here the site count. So $P_s \approx 7 \times 10^{-4}$ to 1.5×10^{-3}, and throwing away the failures would waste 99.9% of the runs.

Amplitude amplification: making a rare success near certain

recipe steps 1, 2

- **Two directions.** The index register reads 0 or not: $\sin\theta\,|\text{success}\rangle + \cos\theta\,|\text{failure}\rangle$, the success part $J|\Psi_0\rangle$ normalized, $\sin\theta = \sqrt{P_s}$; for carbon $\theta \approx 1.5°$–$2.2°$.
- **Two mirrors per round** [Brassard et al. 2002]. Mirror 1, the failure axis: flip the sign of the success part, a flip conditioned on the index reading 0 (7-qubit AND chain, cheap). Mirror 2, the start line: undo the preparation and insertion, flip the sign of all but the all-zero state (AND chain over the 263 system and index qubits), redo them. Two mirrors θ apart turn every arrow by 2θ.
- **Rounds.** After m rounds the arrow is at $(2m+1)\theta$; $m \approx \pi/(4\sqrt{P_s}) = 20$–$29$ lands it near a quarter turn: success with near certainty. More rounds turn it past success, and the chance falls again.
- **The bill:** $1 + 2m = 41$–59 preparations and insertions per circuit, against $1/P_s \approx 680$–$1{,}370$ by trying and discarding.



Drawn for $\theta = 12°$: three rounds reach $84°$. Mirror 2 is the line through the start arrow.

Fewer rounds give many short circuits in place of one deep one (back-up 24).

Everything we can price, added up

recipe steps 1–3

One circuit for $^{12}\mathrm{C}$ on the 4^3 lattice, in the costliest case of the strength band ($m = 29$ rounds):

piece	T gates	share of the 10^9 budget
the energy measurement (walk and readout)	2.17×10^7	2.17%
59 insertions, 508 T each	3.0×10^4	0.003%
29 rounds of reflections, 1,064 T each	3.1×10^4	0.003%
everything priced	$\mathbf{2.18 \times 10^7}$	**2.2%**
41–59 ground-state preparations	unpriced	

- If preparing the ground state were free, carbon would use 2.2% of the 2033 budget.
- The priced parts alone set $\epsilon_l \leq 0.1/(2.18 \times 10^7) = 4.6 \times 10^{-9}$ per hard operation (the error-budget slides).

The allowance: what each preparation may cost

recipe step 1

- **The budget left over:** $10^9 - 2.18 \times 10^7 = 9.78 \times 10^8\ T$, for the preparations.
- **The number of preparations:** $1 + 2m$, run forward or backward inside the amplification; both directions cost the same.
- **So each preparation may cost**

$$\frac{10^9 - 2.18 \times 10^7}{1 + 2m} = 1.7 \times 10^7 \text{ to } 2.4 \times 10^7\ T \qquad (m = 29 \text{ to } 20).$$

  This is the *allowance*: the circuit fits the 2033 budget if a preparation method meets it.
- **For scale:** helium's whole load, 19,683 amplitudes and the spreading, is $2.55 \times 10^6\ T$. Carbon's allowance is 7 to 9 times that.

Three ways to prepare a ground state on the machine

recipe step 1

- **Adiabatic.** Start in the ground state of an easy Hamiltonian H_0, say free nucleons in their lowest orbits, and turn H_0 into H slowly. The adiabatic theorem keeps the system in the ground state if the change is slow compared with the inverse square of the gap to the first excited state. The cost is set by the smallest gap on the path.
- **Variational.** A circuit with adjustable parameters, tuned by a classical optimizer to minimize the measured energy. The depth is yours to choose, but how close the result comes to the true ground state is unknown in advance, and the tuning takes many runs.
- **Projection from a trial state.** Load a cheap trial state whose overlap with the ground state is γ, and filter out the rest by an energy measurement. The cost is about $\lambda/(\gamma\Delta)$ uses of H's block encoding, up to a logarithm, with Δ the gap above the ground state [Lin & Tong, Quantum 4, 372 (2020), Thm. 6]. One use with its reflection is a walk step: 10,548 T ("One walk step, priced").

Why carbon's preparation is still unpriced: our view

recipe step 1

- **The algorithms exist.**
- **Their inputs do not.** Each price needs numbers about carbon *in this model*: the gap along an adiabatic path, or the gap Δ above the ground state and the overlap γ of a trial state we can afford to load. Nobody has computed them for this Hamiltonian on this lattice (whether needing them is circular: back-up 25).
- **A rule of thumb (ours).** At 10,548 T a use, the allowance pays for 1,572 to 2,262 uses of H's block encoding in each preparation, and projection needs about $\lambda/(\gamma\Delta)$. At $\lambda = 21{,}646$ MeV that fits only if $\gamma\Delta \gtrsim 9.6$–13.8 MeV: with an overlap of 0.5, a gap above 19 to 28 MeV. If the budget counted a Toffoli as one hard operation, that would fall by about four: $\gamma\Delta \gtrsim 2.4$–3.5 MeV, a gap above 5 to 7 MeV.
- **So: a grind first, then perhaps invention.** The grind: estimate Δ and γ classically, for example by the imaginary-time projection that lattice EFT already runs for carbon [Epelbaum et al., PRL 106, 192501 (2011)]. If the numbers come out badly, the invention: better trial states, better paths, or a design that prepares the state fewer times.
- Pricing a preparation under this allowance is the open research problem of this calculation, and it is sized: a project for one of you.

The error budget (i): two kinds of error

- **A logical fault** is a gate that went wrong despite error correction, with nothing to flag it. Some faults barely move the answer and some ruin the run; nobody has mapped which for this circuit, so we count every faulted run as lost. **Synthesis error** is a small, known bias in every run, bounded in size.
- **Synthesis:** 0.01 **per circuit,** coherent, shared among the rotations. That is what set 34 T per rotation.
- **Faults:** 0.1 **expected per circuit.** Then at most 10% of runs are lost. A histogram tolerates that if the lost runs scatter over the register's 4,096 values, which we assume. With N_T hard operations in a circuit, $\epsilon_l \leq 0.1/N_T$: 4.6×10^{-9} per hard operation on the priced parts alone, 10^{-10} if the preparation uses up the budget.
- The RFI states no ϵ_l for 2033. At its 2028 figure, 10^{-8}, the priced parts expect 0.22 faults a circuit, twice the budget, and a circuit that uses the whole 2033 budget expects 10. *The application sets the spec the benchmark left open.*

The error budget (ii): what the error-correcting code must deliver

- **Distance.** From hour 1's one-slide arithmetic, $\epsilon_l \sim (p/p_{\rm th})^{(d+1)/2}$: at a physical error rate ten times below threshold, 10^{-10} takes distance $d \approx 19$.
- **Physical qubits per logical one:** about $2d^2 \approx 700$.
- **For this circuit:** the 524-qubit stage needs about 4×10^5 physical qubits for the algorithm's qubits alone.
- So the 2033 machine has to mean, for this calculation, several hundred thousand physical qubits for the algorithm alone, running 10^9 T gates with nine runs in ten free of faults.

The qubit count

register	qubits	stage total
system: one per mode	256	
during the walk: the energy register (12), the prepare's ancillas (43), the reflection's AND chain (42)	97	353
during the amplification: the insertion's index (7) and the zero-state reflection's AND chain over the 263 system and index qubits (261)	268	524

- The peak is 524, plus whatever the preparation needs. Ancillas that are clean at every reflection take turns with the AND chain, so up to 737 are free within the budget's 1,000 qubits. Ancillas the reflection must cover also lengthen the chain, so then only 238.
- Against the budget: fits with room.
- These are *algorithmic* qubits, the ones the circuit names, and we read the budget's 1,000 the same way; the RFI does not say. On the leading architectures a chip needs more, for routing and for factories (back-up 26).

How many runs: one run, one bin

- **One run, one bin.** The response in bin j is the total weight of the final states whose energies fall in it, $S_j = \sum_{f \in j} |\langle f|J|\Psi_0\rangle|^2$. The strength is the total over all bins, $\eta^2 = \sum_j S_j$ ("Applying the current"). A successful circuit returns bin j with probability $p_j = S_j/\eta^2$, the bin's share of the strength. Over N runs the counts are multinomial.
- **Carbon's bins** are 33.2 MeV wide, so a response 200–500 MeV wide (our assumption) fills 7 to 16 bins, and a peak bin holds $p \approx 0.1$ to 0.3 of the runs.
- **To pin a peak bin** of weight $p = 0.15$ to 5% of itself takes $N = (1-p)/(p\,\epsilon^2) \approx 2{,}300$ runs; to pin it to 0.5% of the total strength, $N = p(1-p)/\epsilon^2 \approx 5{,}100$.
- **One histogram is one momentum transfer and one response function.** A response surface, five response functions at five momentum transfers, is 25 histograms: 57,000 to 128,000 circuits.

Wall time: T gates divided by the T rate

- **The model:** wall time = T gates / T rate, on one machine. No roadmap states a T rate, so the rate is our assumption: 10^6 T per second.
- **One circuit:** the priced parts, 2.2×10^7 T, take 22 s; a circuit that uses the whole budget, 10^9 T, takes 17 minutes.
- The 25-histogram set: 2 weeks to a month on the priced parts alone; 2 to 4 years if every preparation spends its whole allowance.
- **The rate is a big lever:** ten times longer at 10^5 T per second, and longer still at the 7×10^3 of the early layout on the T-factory slide [Litinski, arXiv:1808.02892].

The verdict against the 2033 budget

case	the energy measurement	the ground state	algorithmic qubits	standing
^{12}C, 4^3, 23.5 MeV RMS	2.2×10^7 T, 2.2%	41–59 preparations, unpriced; allowance 1.7–2.4 $\times 10^7$ T each	524 + the preparation's	open: fits if a preparation meets its allowance
^{4}He, 3^3, 19.8 MeV RMS	5.3×10^6 T, 0.53%	29–43 loads from a classical table, 0.74–1.1 $\times 10^8$ T	286	fits: 7.9–11.5%
^{4}He, 4^3 (the next lattice), 11.7 MeV RMS	4.4×10^7 T, 4.4%	71–101 loads of 3.9×10^7 T: 2.8–3.9 $\times 10^9$	589	over, $\times$2.8–4.0, through the loads

- **Conventions, ours:** 4 T per Toffoli (counted as one hard operation, helium on the next lattice would fit, at 70–99%); one coherent synthesis budget of 10^{-2} per circuit; 0.1 expected faults per circuit; the strength band $\eta^2 \in [Z/2, Z]$; RMS errors are what the walk delivers.
- ϵ_I: carbon 4.6×10^{-9} on the priced parts and 10^{-10} at a full budget; helium 0.9–1.3 $\times 10^{-9}$; helium on the next lattice 2.5–3.6 $\times 10^{-11}$.
- *The budget constrains one circuit; the runs are a separate axis.*

Argon (i): the same recipe in a bigger box

- **The box.** Forty nucleons, about 7 fm across, need at least the 5^3 lattice, 7.0 fm on a side: 500 modes, one qubit each, as for carbon.
- **The energy measurement** (step 3): 8.26×10^7 T, 8.3% of the budget, at 22.9 MeV RMS. Step 2's insertions and reflections add 0.02%. The preparation (step 1) runs 47 to 65 times in one circuit and is unpriced: to fit the budget, each may cost at most 1.4–2.0 $\times 10^7$ T.
- **The qubits:** 1,018 algorithmic qubits, just past the budget's 1,000: the 510 of the system and index, and 508 ancillas held at once by the AND chain of the amplification's long reflection. Building that AND in blocks, and reusing the ancillas from block to block, would hold 46 instead of 508, for 0.006% of the budget in extra T. That would bring argon back under 1,000 (exercise 4).
- **A bigger box,** 8^3 (11.2 fm): the energy measurement alone is 1.3 times the budget (a third of it, if a Toffoli counted as one hard operation), on 2,166 qubits, and the reflection's AND chain takes the peak to 4,114.

Argon (ii): store positions instead, and the same missing piece

- **The encoding.** Store each nucleon's position instead of each mode's occupation (*first quantization*; arXiv:2507.22814). The register then grows with the nucleon count rather than the box: 498 qubits at 8^3.
- **The cost:** 3.4×10^7 to 3.4×10^8 T per evolution for a $100 \to 10$ MeV window, about 8 times cheaper than the lattice walk in the same box at a matching resolution (2.6×10^9 T at 11.7 MeV RMS), on about a quarter of the register.
- **The same open preparation,** and no classical table either: 8×10^{56} amplitudes at 5^3.
- Argon is where the classical frontier is open too: its inelastic responses "remain unexplored within an ab initio framework" [Acharya et al., arXiv:2410.05962].

Same recipe, bigger box, better encoding, same missing piece.

The RFI, the roadmap and the whitepaper

- **The roadmap people take seriously.** The DOE RFI's two configurations, 2028 (150–250 logical qubits, 10^5 hard ops, $\epsilon_I \leq 10^{-8}$) and 2033 ($\sim 10^3$ qubits, $\sim 10^9$ hard ops), track the public vendor statements: IBM's Blue Jay (2033+, 10^9 gates on $\sim$ 2,000 qubits); Google's 10^6-physical, $\sim 10^3$-logical end state. No public roadmap states $\epsilon_I \lesssim 10^{-9}$ or a T rate.
- **Nothing in this calculation fits 2028**: carbon's energy measurement alone is 220 times its budget, helium's 50 times, and both need an ϵ_I ten to a hundred times below its 10^{-8}. The 2028 configuration is a pipeline-demonstration class, with a different, smaller calculation.
- **Fermilab's whitepaper** responds to the RFI with the laboratory's program for both configurations; read it for how the lab plans to address them. This lecture is an independent calculation against the same budget.
- **What would change the verdict:** a priced preparation under the allowance; a cheaper select; a machine past 10^9.

Reading a resource estimate critically

What to demand of any resource-estimate paper, ours included:

1. **The method.** Time evolution, a walk, or something else; which paper's construction.
2. **The basis and its size.** Modes, lattice, box; what the register holds.
3. **The delivered resolution**, never the bin width.
4. **Which norm sets the cost** (λ, a 1-norm, here) and **per what**: per step, per circuit, per run, per campaign.
5. **Which gates are counted** (T, Toffoli at what rate; Cliffords free?) and the register envelope.
6. **The synthesis error model**: coherent or randomized, and whether the method's theorem covers the circuit.
7. **The fault budget** and the ϵ_I it implies.
8. **The shot model**, and what the histogram's bins assume.
9. What is unpriced. Here: the ground state.

Two of these questions decided answers today: the synthesis error model (whether the tolerance can loosen), and the encoding comparison (back-up 12), whose "winner" depends on what the budget counts.

The code behind every number

- **Every number in hour 2** carries a citation or comes from our code. Where a claim is ours and not the literature's, the slide says so.
- **You can rerun it.** Our tested library, one script and one notebook are released for this school at `github.com/gnperdue/neutrino-school-2026-qc`. The notebook, `price_one_circuit.ipynb`, works hour 2 slide by slide, with the details and the math these slides leave out: read the two side by side. The script, `price_one_circuit.py`, prints carbon's calculation in the slides' order, and its options run helium, argon and the exercises. Both need Python and a few packages. Three side results come from scripts we have not released: the deuteron and triton on the model's Hamiltonian, argon's block AND chain, and argon's comparison with stored positions.
- Hold any resource estimate to the same rule, ours included: a citation or a script for every number, and the method, the basis, the delivered resolution and the gates counted attached to every claim. A T count without those labels is not a number.

Exercises (i): the energy measurement

1. A finer energy register. Carbon on 4^3 with a 13-bit and a 14-bit register: the bin width, the walk steps, the energy measurement's T.

- *How to start:* the bin is $2\pi\lambda/2^{W_q}$; the steps are 2^{W_q-1}; a step stays close to 10,548 T. The readout grows too, since the window state has 2^{W_q} amplitudes.
- *What to look for:* at what register size the energy measurement alone passes the budget, and what resolution that buys.

2. A bigger box. Carbon on 5^3 (125 sites): the term count, the one-norm, the T per step.

- *How to start:* the 38 terms per site and the 338.2 MeV of λ per site do not change with the box, so count again. The select's Toffolis follow the terms, and the reflection changes only through the register widths. The prepare changes too: unlike 64, the odd 125 takes a round of amplification to put in equal superposition.
- *What to look for:* the two ways the box enters, through Γ in each step and through λ in the steps a fixed resolution needs. How fast does the energy measurement grow with the number of sites?

Exercises (ii): price a preparation

3. The open problem, as an exercise. Take a preparation method you know: adiabatic from free nucleons, a variational circuit, or projection from a Slater determinant. Price it against carbon's allowance of $1.7\text{–}2.4 \times 10^7$ T per preparation.

- *How to start:* write the method's cost in uses of H's block encoding or in steps of time evolution; the allowance pays for 1,572 to 2,262 uses. List every number about the nucleus that the cost depends on: a gap, an overlap, the length of a path, the number of parameters and of optimizer runs.
- *Then:* for each number, say how you would get it. From a classical calculation? Which one, and does it exist for this lattice? From a smaller quantum calculation?
- *What to look for:* which unknown dominates, and whether the answer is "fits", "does not fit" or "we cannot say yet". The last is a real result, if you can say what would decide it.

Exercises (iii): the register and the table

4. Argon's register. The lattice version of argon passes 1,000 qubits at 5^3 through one reflection's AND chain. Design the block version and price its extra T.

- *How to start:* an AND chain over k qubits holds $k-1$ ancillas at once. Compute the AND of blocks of qubits first, then the AND of the blocks; ancillas can be cleared and reused between blocks.
- *What to look for:* the trade between ancillas and ANDs, and whether 462 qubits are worth 0.006% of the budget in T.

5. Why 3×10^{18} amplitudes is not a loadable object. Which symmetry of the box reduces it, by how much, and is that enough?

- *How to start:* the translations (helium's trick), then the lattice's rotations and reflections, and the spin and isospin symmetries of the Hamiltonian.
- *What to look for:* the size of the table after every symmetry you can use, against what a table of D amplitudes costs to load: fewer than $(2L+3)D$ Toffolis.

Where to go from here (i): learning the field

Quantum computing, from zero:

- Nielsen & Chuang, *Quantum Computation and Quantum Information*: the standard text; Chs. 1–4 cover everything the on-ramp compressed.
- Preskill, Caltech Ph219 lecture notes (online): the circuit model and error correction, physicist-friendly.
- de Wolf, *Quantum Computing: Lecture Notes*, arXiv:1907.09415: compact and rigorous.

Fault tolerance and the T-gate economy:

- Fowler, Mariantoni, Martinis & Cleland, the surface-code review, arXiv:1208.0928; Bravyi & Kitaev, magic-state distillation, PRA 71, 022316 (2005).
- Litinski, arXiv:1808.02892: Clifford gates commuted away, layouts and factories; Gidney, Quantum 2, 74 (2018): the 4-T AND.
- Girvin, arXiv:2111.08894 and Devitt, Munro & Nemoto, arXiv:0905.2794: our two suggested QEC on-ramps; a primer and a reading guide sit in the back-up.

Where to go from here (ii): the papers this lecture stands on

- The first proposal [Roggero, Li, Carlson, Gupta & Perdue, arXiv:1911.06368] sampled a real-time correlator, smearing the δ with a Gaussian; here we read the energy directly, by phase estimation of a quantum walk [ibid., App. C; arXiv:2004.04889].
- The Hamiltonian and the walk's idea: Roggero et al., arXiv:1911.06368; the contacts' fit: Rokash et al., arXiv:1308.3386.
- The walk's circuits: Babbush et al., arXiv:1805.03662; the walk as a rotation: Low & Chuang, Quantum 3, 163 (2019).
- The ground-state load: Fomichev et al., arXiv:2310.18410.
- The insertion: Roggero, Gu, Baroni & Papenbrock, arXiv:2009.13485.
- The synthesis cost: Watson et al., arXiv:2312.05344; Bocharov, Roetteler & Svore, arXiv:1404.5320; Campbell, arXiv:1612.02689.
- Ground-state preparation by projection: Lin & Tong, Quantum 4, 372 (2020).
- The classical benchmark and the five response functions: Lovato et al., arXiv:2003.07710.
- Argon in first quantization: arXiv:2507.22814.
- And our resource-estimate code (the code slide).

Back-up slides

Reference material: the glossary, a worked example on helium, spare derivations and a QEC primer.

Glossary I

Working definitions, in order of first appearance.

- Inclusive response: the response summed over every hadronic final state, what an experiment that sees only the lepton measures.
- Species: the four kinds of nucleon on a site, $p\uparrow$, $p\downarrow$, $n\uparrow$, $n\downarrow$. Mode: one site and one species.
- Amplitude: the complex number a quantum state assigns to one configuration; its squared magnitude is a probability.
- Qubit: any two-level quantum system. Register: a named group of qubits that holds one quantity.
- Gate: a unitary on one or two qubits. CNOT: flips the target when the control is $|1\rangle$. H, Hadamard: makes the equal superpositions. S: a phase i on $|1\rangle$. T gate: $R_z(\pi/4)$, the expensive gate of a fault-tolerant machine.
- Pauli string: a product of X, Y and Z across qubits. Clifford gate: one that turns every Pauli string into another; Clifford circuits alone can be simulated classically (Gottesman–Knill).
- NISQ: today's noisy machines, without error correction.
- Logical qubit: a qubit encoded in many physical ones and corrected as the program runs. ϵ_l: its error per operation. Distance d: the number of physical errors that can fool the code. Syndrome: the parity outcomes that locate an error.
- Pauli frame: the classical record that replaces running Pauli gates. Magic state: $|T\rangle = T|+\rangle$, consumed to make one T gate. Distillation: many noisy magic states in, a cleaner one out. T factory: the part of the chip that distills. T rate: magic states delivered per second.
- Hard operations: T gates and Toffolis; the easy ones are the Clifford gates and measurements.

Glossary II

- Circuit, run: one execution of prepare, insert and measure, ending in one energy bin. The 2033 budget: 10^9 hard operations per circuit on about 10^3 logical qubits.
- Toffoli: the controlled-controlled-NOT. AND: a Toffoli onto a fresh $|0\rangle$, 4 T, erased for free. AND chain: ANDs in a row, testing whether many qubits are all $|1\rangle$.
- Ancilla: a scratch qubit that starts and ends in $|0\rangle$. Uncomputing: running a computation backward to clear its ancillas.
- Hopping: the term of H that moves a nucleon to a neighboring site. Contacts: the terms that count two (C_0) or three (D_0) nucleons on one site.
- Second quantization: storing each mode's occupation, one qubit per mode; this lecture's choice. First quantization: storing each nucleon's position, the mode it sits in.
- Jordan–Wigner: the map from fermion operators to Pauli strings, with a Z on every mode a hop passes over.
- Γ: the number of Pauli strings in H. λ: the sum of their coefficients' magnitudes, the one-norm.
- Block encoding: a unitary with H/λ in the corner where its ancillas read 0. PREPARE: writes the coefficients onto an ancilla register. SELECT: applies the term the ancilla register names. REFLECT: flips the sign of every ancilla state but $|0\rangle$.
- Oracle: a subroutine an algorithm calls and counts per call; PREPARE and SELECT are the block encoding's oracles.
- Alias sampling: PREPARE's game of chance: a fair die over the entries, then a biased coin that keeps the entry or returns its alternate. Unary iteration: SELECT's tree of ANDs.

Glossary III

- Walk: the block encoding and the reflection, repeated; it turns each energy eigenstate by an angle $\arccos(E/\lambda)$.
- Phase kickback: a controlled unitary moving its eigenphase onto the control. Phase estimation: reading an eigenphase onto an energy register of W_q bits.
- Bin: one value of the energy register, $2\pi\lambda/2^{W_q}$ wide at mid-band. Window state: the register's starting state that narrows the lineshape. Delivered error: the RMS error a run actually achieves.
- Synthesis: building a rotation from T and Clifford gates. Repeat until success: synthesis that retries on failure. Coherent error: one that adds linearly; randomized errors add in squares.
- Load: writing a classically known state onto the system register. Pinning: holding one nucleon at the origin to shrink the table. Translation symmetrization: spreading the pinned state over the lattice.
- Insertion: applying the current. Strength η^2: the response's total weight at one $\vec{q}$. P_s: the insertion's success probability.
- Amplitude amplification: rounds of two reflections that turn a rare success into a near-certain one. Allowance: the T each ground-state preparation may cost for the circuit to fit the budget.
- Adiabatic, variational, projection: the three families of ground-state preparation. Overlap γ: of a trial state with the ground state. Gap Δ: from the ground state to the first excited state.
- Routing: the space a chip needs between logical qubits. Lattice surgery: merging surface-code patches to make them interact.

The bookkeeping wall, in numbers you can feel

- A state of N two-level systems has 2^N amplitudes. Store one byte each: 50 electrons is $2^{50} \approx 10^{15}$ bytes, a petabyte, the memory of a world-class supercomputer. 100 electrons is 2^{100}: $\sim 10^{15}$ such supercomputers.
- Moore's law doubles memory every couple of years, which buys *one more electron* per doubling. Dennard scaling ended two decades ago, so a chip cannot even be switched on all at once.
- Nature simulates itself for free: a test tube of the molecule *is* a quantum simulator, and a superb one. But it answers one question, it is not programmable, and it needs the actual stuff. What we want is a programmable quantum system.
- The nuclear version of the wall is on "The nuclear wall, counted": twelve nucleons on hour 2's 64-site lattice, 3×10^{18} amplitudes.

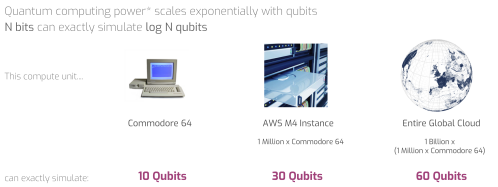


Figure: M. Sohaib Alam.

Exact classical simulation of n qubits needs 2^n numbers: ten qubits on a home computer of the 1980s, thirty on a cloud server, sixty on the whole cloud.

The T gate by teleportation, multiplied out

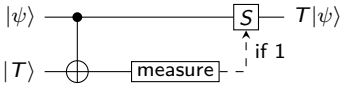
Write $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ and $\omega = e^{i\pi/4}$, so that $|T\rangle = (|0\rangle + \omega|1\rangle)/\sqrt{2}$.

Start, in the order $|00\rangle$, $|01\rangle$, $|10\rangle$, $|11\rangle$, then the **CNOT**, which swaps the last two amplitudes:

$$|\psi\rangle|T\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}\alpha\\ \alpha\omega\\ \beta\\ \beta\omega\end{pmatrix}, \quad \frac{1}{\sqrt{2}}\begin{pmatrix}1&0&0&0\\0&1&0&0\\0&0&0&1\\0&0&1&0\end{pmatrix}\begin{pmatrix}\alpha\\ \alpha\omega\\ \beta\\ \beta\omega\end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix}\alpha\\ \alpha\omega\\ \beta\omega\\ \beta\end{pmatrix}.$$

Measure the second qubit. Each outcome has probability $1/2$, and the first qubit is left in the entries that match it:

- **Reads** 0 (entries 1 and 3): $(\alpha,\ \omega\beta)^{\mathsf{T}} = T|\psi\rangle$. Done.
- **Reads** 1 (entries 2 and 4): $(\omega\alpha,\ \beta)^{\mathsf{T}} = \omega\,(\alpha,\ \omega^{-1}\beta)^{\mathsf{T}} = \omega\, T^\dagger|\psi\rangle$, the wrong way by a known amount. Since $S = T^2$, the S gate gives $\omega\, T|\psi\rangle$, which is $T|\psi\rangle$ up to an overall phase.



The measurement is random, and either outcome is fixed by at most a Clifford gate. The CNOT, the measurement and the S are all easy, so the whole T gate came out of the one magic state.

The arithmetic of distillation

- **Distillation:** 15 noisy copies in, one cleaner copy out, with error $\epsilon_{\text{out}} \approx 35\,\epsilon^3$ [Bravyi & Kitaev, PRA 71, 022316 (2005), Eq. 38].
- From a physical error of 10^{-3}, one round of 15-to-1 reaches 3.5×10^{-8}, short of the 10^{-9} to 10^{-10} that hour 2's circuit needs. A second round reaches about 10^{-21}, and eats $15^2 = 225$ noisy states per clean one. Real factories are tuned in between, but each T gate occupies many physical qubits for a while.

The five response functions

The inclusive charged-current cross section, with $\vec{q}$ along z [Lovato et al., PRX 10, 031068 (2020), Eqs. 2 and 3]:

$$\frac{d\sigma}{dT_\ell\, d\cos\theta_\ell} = \frac{G_F^2 \cos^2\theta_c\, |\vec{k}_\ell|}{4\pi E_\nu} \Big[v_{00} R_{00} - v_{0z} R_{0z} + v_{zz} R_{zz} + v_{xx} R_{xx} \mp v_{xy} R_{xy} \Big],$$

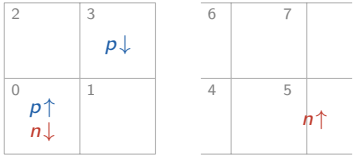
with $-$ for neutrinos and $+$ for antineutrinos, and

$$R_{\alpha\beta}(\vec{q}, q_0) = \sum_f \langle f | J^\alpha(\vec{q}) | \Psi_0 \rangle \langle f | J^\beta(\vec{q}) | \Psi_0 \rangle^* \, \delta(q_0 - E_f + E_0).$$

- The $v_{\alpha\beta}$ are lepton kinematics, known exactly; the $R_{\alpha\beta}$ are the nuclear physics. 00 is the charge, $0z$ and zz the current along $\vec{q}$, xx the transverse current, and xy the interference of the vector and axial currents, whose sign flips between ν and $\bar{\nu}$.
- **On the lattice,** $|\vec{q}\,|$ takes the box's momenta, $2\pi/L$ times integer vectors: 221 MeV for $(1,0,0)$, up to 767 MeV for $(2,2,2)$ on the 4^3 lattice.
- **This lecture prices one representative current,** the proton density. Each response function needs its own histograms (00, zz and xx directly, $0z$ and xy through one combined current each), which is where the factor of five in the run count comes from.

Helium on a $2 \times 2 \times 2$ lattice (i): both encodings, bit by bit

- **A toy small enough to write out:** $^4\mathrm{He}$, one nucleon of each species, on 8 sites numbered 0 to 7 ($x + 2y + 4z$). The 32 modes line up species by species: $p\uparrow$ on sites 0–7, then $p\downarrow$, $n\uparrow$, $n\downarrow$.
- **Configurations:** each species puts its one nucleon on one of 8 sites, $8^4 = 4{,}096$ in all (carbon on 4^3: $41{,}664^4 \approx 3 \times 10^{18}$).
- **The example** (figure): $p\uparrow$ on site 0, $p\downarrow$ on 3, $n\uparrow$ on 5, $n\downarrow$ on 0.



The two planes of the toy lattice, with the example's four nucleons.

encoding	the example's basis state	qubits
occupations	10000000 00010000 00000100 10000000 ($p\uparrow$ | $p\downarrow$ | $n\uparrow$ | $n\downarrow$)	32
positions	00000 01011 10101 11000 (modes 0, 11, 21, 24)	$4 \times 5 = 20$

Of the $2^{32} \approx 4.3 \times 10^9$ occupation strings, 4,096 are helium. Positions hold the same 4,096 states, each the antisymmetric sum of one string's $4! = 24$ orderings, among $2^{20} \approx 1.0 \times 10^6$ strings.

Helium on a $2 \times 2 \times 2$ lattice (ii): one hop, as Pauli strings

In this order a hop moves a nucleon inside its species' block of 8 modes. The $p\uparrow$ nucleon's three hops from site 0, by the Jordan–Wigner rule:

hop	modes between	$a_i^\dagger a_j + a_j^\dagger a_i$
site $0 \to 1$, along x	none	$\frac{1}{2}(X_0X_1 + Y_0Y_1)$
site $0 \to 2$, along y	1	$\frac{1}{2}(X_0Z_1X_2 + Y_0Z_1Y_2)$
site $0 \to 4$, along z	1, 2, 3	$\frac{1}{2}(X_0Z_1Z_2Z_3X_4 + Y_0Z_1Z_2Z_3Y_4)$

- **Each Z reads one mode the hop passes over:** $+1$ if it is empty, -1 if it is full. Those modes are other sites of the same species.
- **Helium:** one nucleon per species, so the modes passed over are always empty and every sign is $+$. On the example, the hop along x turns the $p\uparrow$ block 10000000 into 01000000, with amplitude $-t$, and leaves the other three blocks alone.
- **Carbon:** three nucleons per species, so a hop can pass over another nucleon of its species and pick up the -1 of exchanging two identical fermions. The Z's are how the qubits remember it.

Helium on a $2 \times 2 \times 2$ lattice (iii): a contact, as Z strings

In the example, site 0 holds the $p\uparrow$ (mode 0) and the $n\downarrow$ (mode 24). The two-body contact counts that pair. With $n = (1 - Z)/2$ on each mode,

$$C_0\, n_0\, n_{24} = \tfrac{C_0}{4}\left(1 - Z_0 - Z_{24} + Z_0 Z_{24}\right).$$

modes 0 and 24	Z_0	Z_{24}	$\frac{C_0}{4}(1 - Z_0 - Z_{24} + Z_0 Z_{24})$
both empty	$+1$	$+1$	0
one full	-1	$+1$	0
both full	-1	-1	$C_0 = -98$ MeV

- A constant, two single Z's and one ZZ together switch the contact on only when both modes are full.
- Summed over the 6 pairs and 4 triples of species at a site, these pieces collect into one site's 4 single Z's, 6 ZZ's and 4 ZZZ's, with the coefficients on "One site's 38 terms". The constants add up to one number, and it is dropped.

Two encodings of the same Hamiltonian, three numbers apart

For time evolution by product formulas, a fermionic encoding maps occupation numbers onto qubits. Jordan–Wigner's long strings cost each hop two-qubit depth; two standard local encodings avoid them, and for this Hamiltonian they differ in exactly three numbers [Watson et al., arXiv:2312.05344, Tabs. 10–11]:

encoding	qubits per site	hopping layer two-qubit depth, all sites	contact layer two-qubit depth, all sites
Verstraete–Cirac	6 (4 system + 2 auxiliary)	512	8
stacked compact	10 (about 2.5 per mode)	60	8

- The rotation counts and the T-count formula are *identical* under both. Only width and hopping depth move, so this is a clean controlled comparison [Watson et al., arXiv:2312.05344, Tabs. 10–12].
- Stacked compact buys an $8.5\times$ shallower hopping layer ($512 \to 60$ two-qubit layers) and pays 4 more qubits per site.
- *Before you read the next slide: which would you pick?*

"Which encoding is better?" is a malformed question

Same two encodings, same Hamiltonian, evaluated against the two configurations' budgets. Widths are the system register, (qubits/site) $\times L^3$; adding a few tens of ancillas changes neither verdict:

	3^3 lattice (27 sites)		4^3 lattice (64 sites)	
Encoding	width	vs. the 2028 budget	width	vs. the 2033 budget
Verstraete–Cirac	162 LQ	fits (150–250)	384 LQ	fits ($\gtrsim 10^3$)
stacked compact	270 LQ	over	640 LQ	fits

- On the 3^3 lattice only Verstraete–Cirac fits the 2028 register envelope. Width binds there, and width decides.
- The 8.5× is two-qubit depth, built from Clifford gates. Both budgets count hard ops, and so does our wall-time model; the T count per step is identical under both encodings. So the shallower layer buys nothing on either machine: at 2033 both fit, they tie on everything the budget measures, and Verstraete–Cirac still uses fewer qubits.
- *Before you bank a saving, check that your budget counts it.* The 8.5× pays only where two-qubit depth is the cost: on hardware without error correction, where every layer adds error, or on a fault-tolerant machine whose run time is set by Clifford layers rather than by T supply. That is the habit this lecture is really trying to teach: *name the binding constraint, and what it counts, before comparing anything.*
- Corollary worth internalizing: a resource estimate quoted without its register envelope, or without saying which gates it counts, is not comparable to another one. The closing matter collects the ways this goes wrong in print.

The whole product-formula trade-off is one inequality

Split $H = \sum_j H_j$ and approximate e^{-iHt} by r repetitions of a product formula. First order carries $\epsilon_{\text{step}}(\tau) \sim K\tau^2$, with K set by the commutators $\|[H_j, H_k]\|$. Choose r by:

$$r \cdot \epsilon_{\text{step}}(t/r) \;\leq\; \epsilon_{\text{budget}}$$

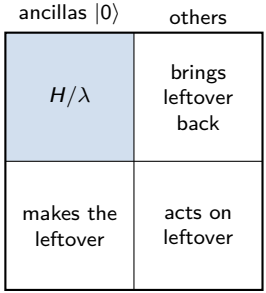
- Larger r: smaller steps, smaller total error, *more gates*. That is the entire tension; the rest is bookkeeping on it.
- For $K\tau^2$ the total is Kt^2/r, so r grows like t^2. Doubling the evolution time quadruples the cost at fixed accuracy. Higher-order formulas trade a larger constant for slower growth.
- *The physics enters only through K:* the EFT order, the lattice size, the commutator bounds.
- Hour 2 never makes this error. The walk applies the block encoding of H exactly, with no product formula; its errors are the rotations' synthesis and the energy register's finite bits.

Block encoding: what the other three blocks hold

None of the starred blocks of Block encoding (iv) is zero: B keeps lengths, so what H/λ does not keep, they must carry.

- **Below H/λ: it makes the leftover.** It sends $|0\rangle|\psi\rangle$ to $s\,|\perp\rangle$, with $s^2 = 1 - \|H|\psi\rangle\|^2/\lambda^2$. It would vanish only if H/λ were itself unitary: an H of one Pauli string, for example.
- **Right of H/λ: it brings leftover back.** B is unitary and $B^2 = I$, so $B^\dagger = B$, and this block is the conjugate transpose of the one below H/λ.
- **Bottom right: it acts on what starts as leftover.** It would vanish only if H were zero.
- **One energy at a time.** In the plane of $|0\rangle|\psi_E\rangle$ and its leftover $|\perp\rangle$, B is the mirror of "Why the walk encodes the energy":

$$B = \begin{pmatrix} E/\lambda & s \\ s & -E/\lambda \end{pmatrix}, \qquad s = \sqrt{1 - E^2/\lambda^2} = \sin\theta_E.$$



The picture of Block encoding (iv), its stars filled in. "Non-zero" is block by block: a block may still hold many zero entries.

The walk and phase estimation: the high-level view

- **One long circuit.** One step is four boxes: PREPARE, SELECT, PREPARE undone and REFLECT (slide 43; priced on slides 44 to 46). The circuit repeats it, unmeasured, 2^{W_q-1} times: 2,048 for $W_q = 12$.
- **A step is a turn, and the walk has no target.** Every step turns an eigenstate of energy E by the same angle θ_E, with $\cos\theta_E = E/\lambda$ (slide 47). Nothing has to arrive: more steps read the angle more finely.
- **The control writes the angle.** A step touches only the ancillas and the system, so repeating it writes nothing into the energy register. Each of the register's W_q qubits *controls* a block of steps, bit k driving 2^k of them, and the block's phase kicks back onto that qubit (slide 48). An inverse Fourier transform turns the W_q phases into the angle written in binary, and an ordinary measurement reads it (slide 49).
- **The factor of two.** Babbush et al. run the step forward on one branch of the control qubit and backward on the other, so the branches part by $2\theta_E$ a step instead of θ_E: $2^{W_q} - 1$ steps become 2^{W_q-1} (slide 49).
- **All in the final state, and we have to sample it.** The current leaves the system in a superposition of energy eigenstates with weights $|\langle f|J|\Psi_0\rangle|^2$. Phase estimation writes each one's energy bin beside it in the register. If we could cheat nature and read a wave function, we would have the response at once; a measurement returns one bin, in proportion to that weight. One run is one sample, and many runs fill the histogram (slide 50).
- **Two readings, one energy.** The walk's eigenvalues are $e^{\pm i\theta_E}$, and the start state is an equal mix of the two eigenstates. The register reads $+\theta_E$ or $-\theta_E$, half the time each, and both give $E = \lambda\cos\theta_E$.

Why the walk needs REFLECT: B alone bounces back

A bit of the energy register never receives a step. Every step acts on the ancillas and the system; bit k only switches on a block of 2^k steps, run back to back (the textbook series of slide 49). Compare the blocks:

	the step is B alone	the step is $W = RB$, the walk
bit 0's block	B	W: a turn by θ_E
bit 1's block	$B^2 = I$	W^2: a turn by $2\theta_E$
bit k's block, $k \geq 1$	$B^{2^k} = I$	W^{2^k}: a turn by $2^k\theta_E$
the register reads	one coin flip	θ_E in binary, to W_q bits

- **The coin flip.** With B alone, bit 0 reads 0 with probability $(1 + E/\lambda)/2$: the energy is in the odds, never in a digit. Many runs give the average energy of the superposition the current leaves, and no histogram.
- **Where the energy hides in B.** A mirror has eigenvalues $+1$ and -1 whatever the energy. B carries the energy in where its mirror line points, $\theta_E/2$ from the start (slide 47). Phase estimation reads how fast something turns; it cannot read where a line points.
- **What REFLECT adds.** A second, fixed mirror. Two mirrors make a rotation by twice the angle between them: every step turns by θ_E, the turns add, and every bit gets its digit.

Why phase estimation is expensive, and what is not expensive about it

- **Resolution is bought with walk steps, and λ is the exchange rate.** The bins are $2\pi\lambda/2^{W_q}$ wide and the walk is 2^{W_q-1} steps, so the steps number $\pi\lambda$ over the bin width: for carbon, $\pi \times 21{,}646/33.2 = 2{,}048$. Halve the bin, double the circuit: linear in $1/\Delta E$, so exponential in the number of digits. Twelve bits sounds like nothing; it is two thousand block encodings.
- **The price of resolution, not waste in the method.** One walk step is worth about $1/\lambda$ of evolution time [Low & Chuang, Quantum 3, 163 (2019)], so $\lambda/\Delta E$ steps is the time $1/\Delta E$ the uncertainty principle demands to resolve ΔE. That is the Heisenberg limit: no method resolves an energy with fewer than order $\lambda/\Delta E$ uses of the block encoding. The levers are λ and the price of a step.
- **The steps are coherent.** All 2,048 run inside one circuit, controlled by the energy register, with nothing measured between them. Resolution sets a depth, and a gate error of one in a thousand fails long before the end: this is what makes phase estimation a fault-tolerant algorithm, and what "it is expensive, use a variational method" meant.
- **Not the register.** The inverse Fourier transform and the window state cost 1.4×10^5 T, 0.014% of the budget: a rounding error next to the walk.
- **Runs multiply it.** One run returns one bin ("one run, one bin"), and the response needs many.
- And still the cheap part. For carbon the energy measurement is 2.2% of the 10^9 budget; in helium's priced circuit the walk is 5–7% and the ground-state loads 93–95%. Expensive in absolute terms, and the smaller bill.

Randomized synthesis: a looser tolerance at the same price

- **The idea.** For each angle build two approximations, one over-rotated and one under-rotated, and pick one at random in every run. The errors no longer line up: a mixed rotation's error is at most $5\epsilon^2$ [Campbell, arXiv:1612.02689, Thm. 2] instead of ϵ, and N of them add to $5N\epsilon^2$.
- **The tolerance loosens.** With the same budget, $5N\epsilon^2 \leq 10^{-2}$ gives $\epsilon = 4.0 \times 10^{-4}$ for carbon's 12,420 rotations, against 8×10^{-7} when the errors add linearly.
- **The price does not.** The theorem covers deterministic synthesis, a fixed Clifford-and-T sequence for each approximation, such as Ross–Selinger's [arXiv:1403.2975] at about $3\log_2(1/\epsilon)$: 34 T at 4.0×10^{-4}, the same as repeat until success at 8×10^{-7}. Repeat until success is not a fixed sequence, so its $1.15\log_2(1/\epsilon)$ at the loose ϵ is not covered. The loose tolerance and the cheap method do not combine.
- **What we do.** Price with the errors adding linearly, by repeat until success, at 34 T: a claim that a circuit fits is safest under the model that adds errors linearly. The notebook's rotation cells give both.

The strength η^2: what it is, and why we take $[Z/2,\ Z]$

- **What it is.** $\eta^2 = \|J|\Psi_0\rangle\|^2 = \sum_f |\langle f|J|\Psi_0\rangle|^2$: the squared length of the state the current leaves, and so the response added up over every energy transfer at this $\vec{q}$: the area under the histogram.
- **A property of the nucleus, not a choice of ours.** The Hamiltonian, the nucleus and $\vec{q}$ fix it, and the machine would measure it: Λ_J^2 times the fraction of insertions that succeed (slide 60). For carbon we have no value, since computing it takes the ground state. The band is our estimate.
- **What it is made of.** For the proton density, $\eta^2 = Z + \sum_{i\neq j}\langle e^{i\vec{q}\cdot(\vec{x}_i-\vec{x}_j)}\rangle$: each proton on its own, plus the interference between every pair.
- **Two limits.** At $\vec{q}=0$ the protons add in step: $\eta^2 = Z^2$, all of it the *elastic line*, the nucleus recoiling whole in its ground state. Well above the Fermi momentum the pair terms average away: $\eta^2 \to Z$, the proton sum rule.
- **In between,** where the lattice's momenta sit, Pauli blocking holds the strength without the elastic line below Z. A free Fermi gas has $0.69\,Z$ at $q = k_F$ and Z from $2k_F$ on; the lattice's smallest momentum, 221 MeV, is about carbon's k_F [221 ± 5 MeV, Moniz et al., PRL 26, 445 (1971)]. We take $[Z/2,\ Z]$.
- **The weak end is the safe side.** A weaker response costs more: the rounds of amplification go as $1/\eta$, 20 at $\eta^2 = Z$ and 29 at $Z/2$, so 41 to 59 preparations (slide 61). The elastic line, left out of the band, only adds strength. Below the band the cost keeps rising: at $\eta^2 = 1$ the allowance would be 1.7 times smaller.

A lever nobody has priced: normalize the current at Z

recipe steps 1, 2

- **Why success is rare.** The current's block encoding divides by its one-norm, $\Lambda_J = M = 64$, the number of sites. But on carbon's states the current only ever acts on $Z = 6$ protons: there, $\|\rho_{\vec{q}}\| \leq Z$.
- **The lever:** a block encoding of the current normalized at Z instead of M would raise P_s by $(M/Z)^2 = (64/6)^2 \approx 114$.
- **What it would buy:** the rounds fall from 20–29 to 1–2, the preparations per circuit from 41–59 to 3–5, and the allowance per preparation rises to $2.0\text{–}3.3 \times 10^8$ T.
- Nobody has built or priced that block encoding (why: two slides on). The bonus challenge on the next slide asks what it would do for helium.

Bonus challenge exercise: the lever, for helium

Normalize the current at Z. Helium on 3^3 with the insertion normalized at $Z = 2$ instead of $M = 27$: P_s, the rounds, the loads, the circuit.

- *How to start:* $P_s = \eta^2/\Lambda_J^2$ with $\eta^2 \in [Z/2, Z]$. The rounds are the integer m that brings $(2m+1)\theta$ nearest a quarter turn, with $\sin\theta = \sqrt{P_s}$; for a large P_s the small-angle formula $\pi/(4\sqrt{P_s})$ is not good enough.
- *What to look for:* how much of helium's circuit is the load before and after, and whether the energy measurement becomes the largest cost.
- *Then the hard part:* nobody has built this block encoding. What would it have to do that the one normalized at the site count does not?

The lever explained: why the one-norm is M, and why Z is hard

- **Where the 64 comes from.** The insertion writes the current as a list of 128 terms of size $\frac{1}{2}$, a Pauli Z on each proton mode, and its block encoding divides by the sum of those sizes, $\Lambda_J = 128 \times \frac{1}{2} = 64 = M$. That number is a property of how the operator is written, counted over every site whether a proton sits there or not.
- **Why 6 is the true size.** On a state of $Z = 6$ protons the current is the sum of six phases of modulus one, one per proton, so $\|\rho_{\vec{q}}|\Psi\rangle\| \leq Z$ for every such state, and a block encoding normalized at Z would succeed with probability $P_s = \eta^2/Z^2 \leq 1$. The strength band, our estimate, puts η^2 between $Z/2$ and Z. Dividing by 64^2 in place of 6^2 throws away a factor $(M/Z)^2 \approx 114$ in P_s for nothing.
- **What the lever buys, link by link.** Rounds go as $1/\sqrt{P_s}$, so 114 in probability is about 10.7 in rounds: 20–29 become 1–2; preparations, $1 + 2m$, fall from 41–59 to 3–5; the allowance per preparation, the leftover budget divided by the preparations, rises about twelvefold to $2.0\text{–}3.3 \times 10^8$ T. Nothing else in the circuit changes.
- **Why nobody has built it.** A linear combination of unitaries cannot: its one-norm is fixed by the written terms, not by the state. In first quantization, one position register per proton, the current is a sum of exactly Z unitaries and the normalization Z comes free; but this calculation is second-quantized, one qubit per mode, and the walk's block encoding of H with it. A second-quantized version needs a PREPARE that reads which modes are occupied and puts the index register over those Z alone: a construction of a different kind, with nothing to price yet.

Step 2's trade: many short circuits, or one deep one

- **One dial, the rounds m.** Stop after m rounds and measure the index: the circuit holds $2m+1$ preparations and lands with probability $\sin^2((2m+1)\theta)$; a miss starts over. $m=0$ is trying and discarding, $m=29$ the lecture's design. For $^{12}\mathrm{C}$ at the costlier end of the strength band:

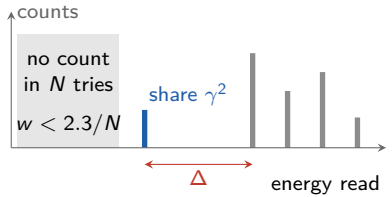
	depth: one circuit			**count:** one success, on average	
rounds m	preparations	T gates, N_T	needs $\epsilon_I \le 0.1/N_T$	circuits	wall time
0	1	3.8×10^{7}	2.6×10^{-9}	1,365	6.3 h
2	5	1.0×10^{8}	9.6×10^{-10}	55	1.3 h
10	21	3.7×10^{8}	2.7×10^{-10}	3.5	20 min
29	59	1.0×10^{9}	1.0×10^{-10}	1.0	17 min

Each preparation at its whole allowance, 1.66×10^{7} T; the energy measurement adds 2.17×10^{7}; 10^{6} T per second.

- **The logical error rate limits the rounds.** A miss starts a new circuit, so faults add up only inside one circuit. A lower error rate allows a deeper circuit, and so more rounds m; a higher one may force $m=0$.
- **Then the wall clock pays.** At $m=0$ a success takes 1,365 circuits and 6.3 hours, against one circuit and 17 minutes. Discarded tries can run in parallel on more machines; the rounds of one circuit cannot. The lecture prices one point on this dial; the right one moves with the machines.

Is projection circular? What a run reads, and when it stops

- **One try reads one line.** Load the trial state and measure its energy with a width δ: about λ/δ uses of H's block encoding. The reading is one eigenstate's energy, picked with its weight, and the register is left in that state.
- **N tries hold γ and Δ.** Add bits to the energy register, each halving δ, as far as the budget allows. If the lowest line stands apart, it is the candidate ground state: its share of the tries is γ^2; the next line is Δ above. To prepare it, keep those tries or amplify, about $1/\gamma$ rounds, hence $\lambda/(\gamma\Delta)$.
- **No run announces a lower line it has not seen.** With none read below the candidate in N tries, a hidden state's weight is below $2.3/N$ at 90% confidence (figure). Certifying the lowest state is hard even for a quantum computer [Kempe et al. 2006].
- **So a run stops on evidence:** that limit; the same line from a different trial state; an outside anchor, the measured binding energy as far as the model is trusted, or imaginary-time projection. With no anchor the result is a limit.



Energies read in N tries of one trial state. Blue: the candidate. Gray: none read; a state of weight w hides with chance $(1-w)^N$, 10% at $w = \ln(10)/N$.

Asking for machine time takes an estimate of γ and Δ, as beam time takes an expected rate: smaller nuclei solved exactly, imaginary-time projection, the first tries.

From the algorithm's qubits to a chip: routing and factories

- **What we counted:** *algorithmic* qubits, the ones the circuit names. On a surface-code chip each is a patch of physical qubits.
- **Routing.** Two logical qubits interact by merging their patches through the space between them (*lattice surgery*). A layout leaves corridors of free patches for that, and wider corridors let more operations run at once [Litinski, arXiv:1808.02892].
- **Factories** (hour 1) sit beside the data and feed it magic states.
- Our allowance: routing multiplies the algorithmic count by about two to three, before the factories. It is our round figure, not a computed layout; a layout for this circuit is one of the open engineering steps.

QEC primer — what limits us to 10^5 hard gates in 2028?

The main deck treats error correction as a given cost model (Clifford cheap, T expensive, an ϵ_I per logical op). The physicist's follow-up — *why those numbers?* — in three constraints [schematic surface-code heuristics; refs on the catch-up slide]:

- Reliability. Each logical op fails with probability ϵ_I; N ops accrue $N\epsilon_I$ expected faults, so a useful circuit needs $N \lesssim (\text{fault budget})/\epsilon_I$. The 2028 configuration is consistent: $10^5 \times 10^{-8} = 10^{-3}$ faults/shot.
- Register. ϵ_I is bought with code distance: $\epsilon_I \sim (p/p_{\text{th}})^{(d+1)/2}$ at physical error rate p. At $p \sim p_{\text{th}}/10$, reaching $\epsilon_I \sim 10^{-8}$ needs $d \sim 15$, i.e. $\sim 2d^2 \approx 450$ physical qubits per logical: 250 LQ $\Rightarrow\sim 10^5$ physical qubits *before* factories.
- Throughput. Clifford gates act cheaply on the surface code; each T gate consumes a *distilled magic state* from factories that occupy much of the machine and emit states at a finite rate — hard ops are serialized fuel, not free instructions.

So the 10^5 is not a fault-accumulation wall ($\epsilon_I = 10^{-8}$ would allow more); it is the physical-qubit and factory budget of a plausible first-generation fault-tolerant machine, which the RFI encodes as a spec.

QEC primer — pooled COVID tests are a parity-check code (Girvin §2.4.1)

Screen 7 samples $S_1, \ldots, S_7$ with only 3 tests: mix portions into 3 *pools* P_1, P_2, P_3, each sample joining a unique subset of pools. If the positivity rate is low (at most one positive sample), the pattern of positive pools identifies it:

- all pools negative $\rightarrow$ no positive sample;
- otherwise, read the pool results as a binary number $(P_3 P_2 P_1)$ — it *is* the index of the positive sample (sample 5 $\rightarrow$ pattern 101).

The pooling map is a matrix — column k is the number k in binary:

$$H = \begin{pmatrix} 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{pmatrix}, \quad P = HS.$$

Three test outcomes — the syndrome — locate one faulty entry among seven *without examining the samples individually*.

That is the entire logic of syndrome decoding. The next slide turns it into an error-correcting code.

QEC primer — the same matrix, mod 2: the Hamming $[7, 4, 3]$ code (Girvin §2.4.2)

- Now let a 7-bit message M carry data, and demand it pass all three parity checks: $HM = 0 \pmod 2$. Exactly $16 = 2^4$ strings qualify — the codewords: $n = 7$ physical bits carrying $k = 4$ logical bits.
- Transmission flips one bit: $M \to M \oplus S$. Because codewords pass every check, the syndrome computes to $H(M \oplus S) = HS$ — *the binary address of the flipped bit*, exactly the pooled-test decoder. Correct it and move on.
- Code distance $d = 3$ (closest codewords differ in three places) $\Rightarrow$ corrects $\lfloor (d-1)/2 \rfloor = 1$ error; the $[n, k, d]$ label $[7, 4, 3]$ summarizes everything.

Why this matters for us: distance is the knob — the quantum surface code buys ϵ_I with d in precisely this spirit (previous slide). And *hold on to H*: it returns, twice, inside the quantum code two slides ahead.

QEC primer — states from operators: the stabilizer formalism (Devitt §VII)

- Flip to the Heisenberg picture: specify a state by the operators that fix it, $K|\psi\rangle = +|\psi\rangle$ — "$|0\rangle$ is the state with $\sigma_z = +1$" rather than a column vector.
- For n qubits, choose k commuting, independent Pauli strings (a stabilizer group): the states they jointly fix form a 2^{n-k}-dimensional subspace. Examples: $|\mathrm{GHZ}_3\rangle \leftrightarrow \{XXX, ZZI, IZZ\}$; the four Bell states are the four sign patterns of $\{\pm XX, \pm ZZ\}$.
- Any Pauli string can be *measured* with one ancilla (H, controlled-K, H, measure): the ± 1 parity outcome projects the register while leaving the encoded superposition untouched. This "operator measurement" is the workhorse of everything below.

A code is then simply a stabilizer group whose fixed subspace is the logical qubit: compact generator lists replace exponentially long state vectors, and preparation/correction circuits are synthesized directly from the list.

QEC primer — error correction with stabilizers: Steane's $[[7, 1, 3]]$ code (Devitt §VIII)

Double brackets = quantum: $n = 7$ physical qubits, $k = 1$ logical qubit, $d = 3$. Six stabilizer generators (note the shapes):

$$K^1 = IIIXXXX, \quad K^2 = XIXIXIX, \quad K^3 = IXXIIXX,$$
$$K^4 = IIIZZZZ, \quad K^5 = ZIZIZIZ, \quad K^6 = IZZIIZZ,$$

fixing a $2^{7-6} = 2$-dimensional code space — one protected qubit.

- Detection = commutation. A Pauli error E either commutes with a generator (measurement outcome $+1$) or anticommutes (-1). Measuring all six yields a syndrome: a single X_i trips a unique pattern of $K^{4,5,6}$; a single Z_i trips $K^{1,2,3}$; $Y_i = iX_iZ_i$ trips both. Correct the addressed qubit — never learning the encoded amplitudes.
- Digitization for free (their §IX): the syndrome measurement *projects* whatever continuous wobble occurred onto a discrete Pauli error — why analog quantum noise is correctable with digital-style codes at all.

QEC primer — which paper should you read?

Girvin, arXiv:2111.08894
Introduction to QEC and Fault Tolerance (Les Houches lecture notes, ∼90 pp).

- Full arc from classical codes and fault tolerance (the COVID pooling is §2.4) through qubit QEC, the toric/surface code, and bosonic/GKP "hardware-efficient" codes.
- Conversational, intuition-first, von-Neumann-to-modern framing. Start here if you want the *story*.

Devitt, Munro & Nemoto, arXiv:0905.2794
Quantum Error Correction for Beginners (review, ∼38 pp).

- Worked examples end to end: 3-, 9-, 7-, 5-qubit codes, the stabilizer formalism (§VII–VIII), error digitization, fault-tolerant circuits, the threshold theorem.
- Compact and operational. Start here if you want the *working formalism* fast.

Read them as a pair: Girvin's pooling matrix H (columns $= 1 \ldots 7$ in binary) is *exactly* the check pattern of Devitt's Steane code — $K^{1\text{–}3}$ and $K^{4\text{–}6}$ are H's three rows, applied once as X-checks and once as Z-checks. One classical idea, used twice, protects a qubit against both flips and phases.